Tugas Pertemuan 9.1 – Latihan DML

Mata Kuliah: Sistem Basis Data - Praktikum (2026)

Prodi: Sarjana Terapan-Teknik Informatika POLBAN

Kelas: 2B

NIM: 251524048

Nama Lengkap: Ghaisan Khoirul Badruzaman

Jawab soal berikut dengan syntax PosgreSQL dan database HR!

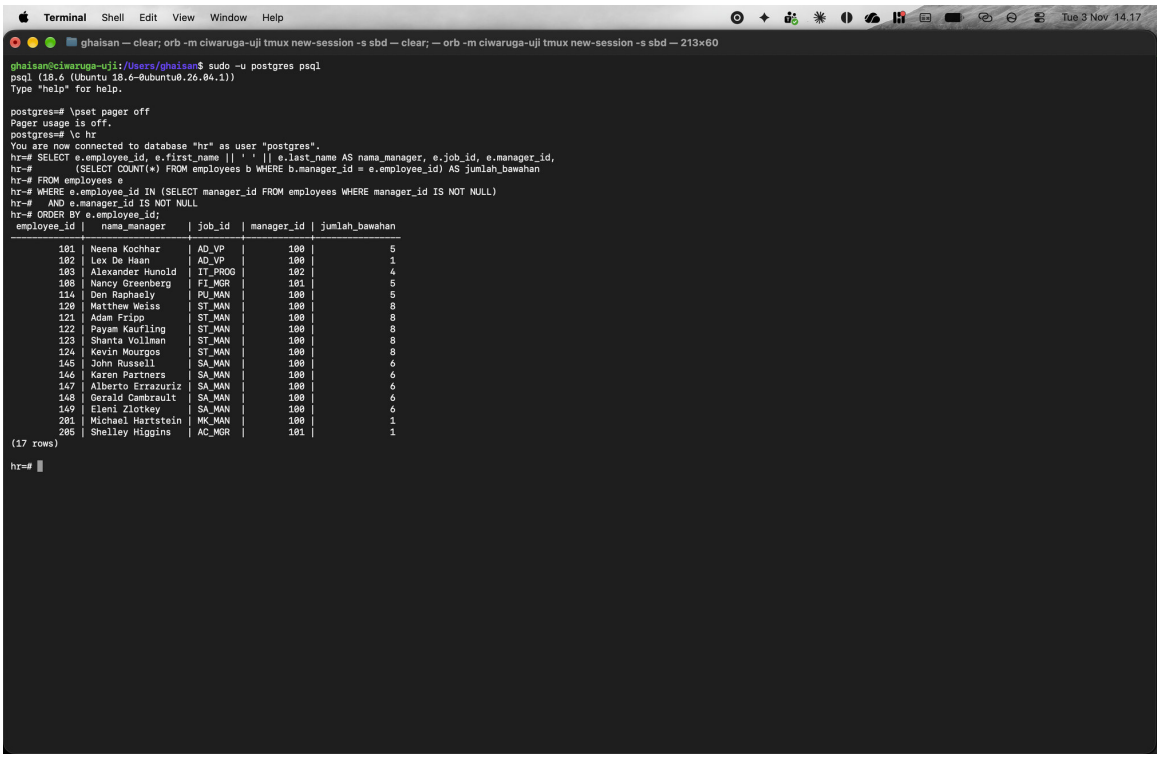
Latihan No. 1

Soal
Cari Data Employee yang merupakan Manager dan bukan merupakan Manager tertinggi!
Syntax

`sudo -u postgres psql`
 	
`\pset pager off`
 	
`\c hr`
 	
`SELECT e.employee_id, e.first_name || ' ' || e.last_name AS nama_manager, e.job_id,`
`  e.manager_id,`
`  (SELECT COUNT(*) FROM employees b WHERE b.manager_id = e.employee_id) AS`
`  jumlah_bawahan`
`FROM employees e`
`WHERE e.employee_id IN (SELECT manager_id FROM employees WHERE manager_id IS NOT NULL)`
`  AND e.manager_id IS NOT NULL`
`ORDER BY e.employee_id;`

Keterangan
Masuk ke psql sebagai user postgres, mematikan pager, lalu pindah ke database hr. Employee dianggap Manager kalau employee_id-nya dipakai sebagai manager_id oleh employee lain, jadi dicek dengan subquery IN (SELECT manager_id FROM employees). Manager tertinggi adalah Steven King (ID 100, President) yang tidak punya atasan, sehingga disaring dengan e.manager_id IS NOT NULL. Kolom jumlah_bawahan dihitung dengan subquery berkorelasi untuk menunjukkan bahwa setiap baris memang punya bawahan. Hasilnya 17 manager, mulai dari Neena Kochhar (5 bawahan) sampai Shelley Higgins (1 bawahan).

Hasil Eksekusi (Screenshot)



```
Terminal  Shell  Edit  View  Window  Help                                                    Tue 3 Nov  14.17
ghaisan — clear; orb -m ciwaruga-uji tmux new-session -s sbd — clear; — orb -m ciwaruga-uji tmux new-session -s sbd — 213×60

ghaisan@ciwaruga-uji:/Users/ghaisan$ sudo -u postgres psql
psql (18.6 (Ubuntu 18.6-0ubuntu0.26.04.1))
Type "help" for help.

postgres=# \pset pager off
Pager usage is off.
postgres=# \c hr
You are now connected to database "hr" as user "postgres".
hr=# SELECT e.employee_id, e.first_name || ' ' || e.last_name AS nama_manager, e.job_id, e.manager_id,
hr-#        (SELECT COUNT(*) FROM employees b WHERE b.manager_id = e.employee_id) AS jumlah_bawahan
hr-# FROM employees e
hr-# WHERE e.employee_id IN (SELECT manager_id FROM employees WHERE manager_id IS NOT NULL)
hr-#   AND e.manager_id IS NOT NULL
hr-# ORDER BY e.employee_id;
 employee_id |   nama_manager    | job_id  | manager_id | jumlah_bawahan
-------------+-------------------+---------+------------+----------------
         101 | Neena Kochhar     | AD_VP   |        100 |              5
         102 | Lex De Haan       | AD_VP   |        100 |              1
         103 | Alexander Hunold  | IT_PROG |        102 |              4
         108 | Nancy Greenberg   | FI_MGR  |        101 |              5
         114 | Den Raphaely      | PU_MAN  |        100 |              5
         120 | Matthew Weiss     | ST_MAN  |        100 |              8
         121 | Adam Fripp        | ST_MAN  |        100 |              8
         122 | Payam Kaufling    | ST_MAN  |        100 |              8
         123 | Shanta Vollman    | ST_MAN  |        100 |              8
         124 | Kevin Mourgos     | ST_MAN  |        100 |              8
         145 | John Russell      | SA_MAN  |        100 |              6
         146 | Karen Partners    | SA_MAN  |        100 |              6
         147 | Alberto Errazuriz | SA_MAN  |        100 |              6
         148 | Gerald Cambrault  | SA_MAN  |        100 |              6
         149 | Eleni Zlotkey     | SA_MAN  |        100 |              6
         201 | Michael Hartstein | MK_MAN  |        100 |              1
         205 | Shelley Higgins   | AC_MGR  |        101 |              1
(17 rows)

hr=#
```

Latihan No. 2

Soal

Tampilkan Data Employee disertai dengan Nama Department, City Location, Country Name, Region Name!

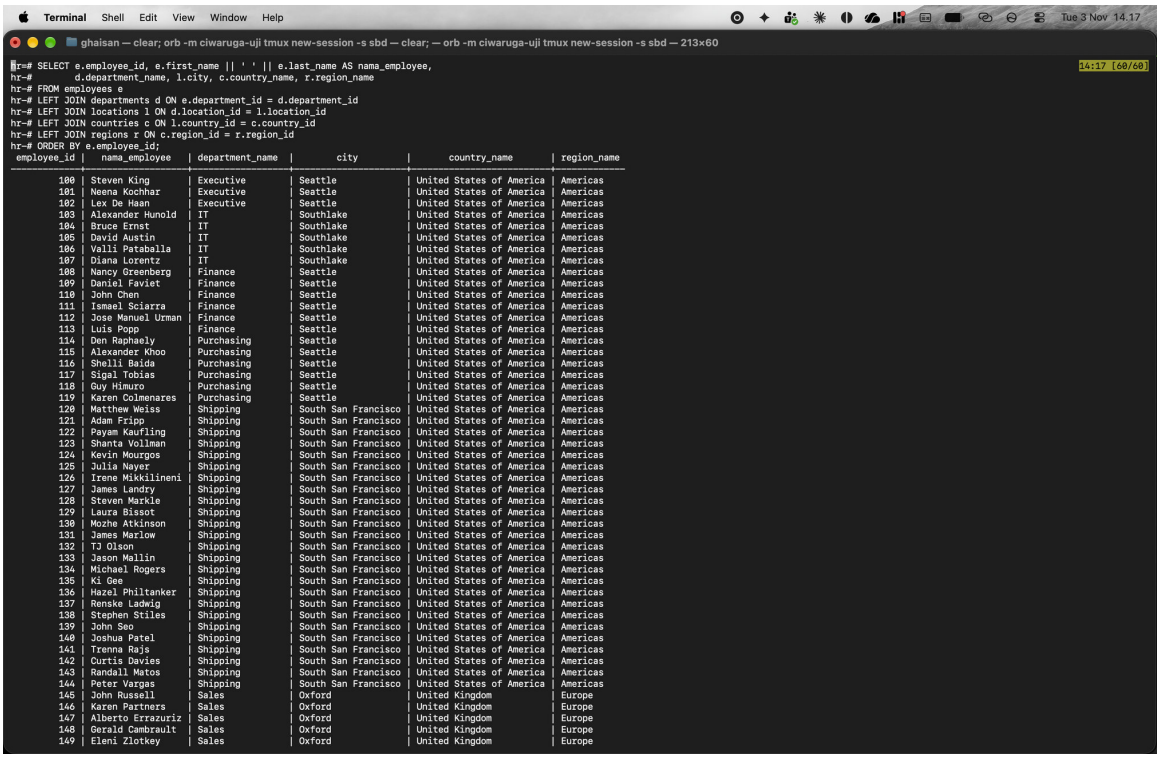
Syntax

```
SELECT e.employee_id, e.first_name || ' ' || e.last_name AS nama_employee,
       d.department_name, l.city, c.country_name, r.region_name
FROM employees e
LEFT JOIN departments d ON e.department_id = d.department_id
LEFT JOIN locations l ON d.location_id = l.location_id
LEFT JOIN countries c ON l.country_id = c.country_id
LEFT JOIN regions r ON c.region_id = r.region_id
ORDER BY e.employee_id;
```

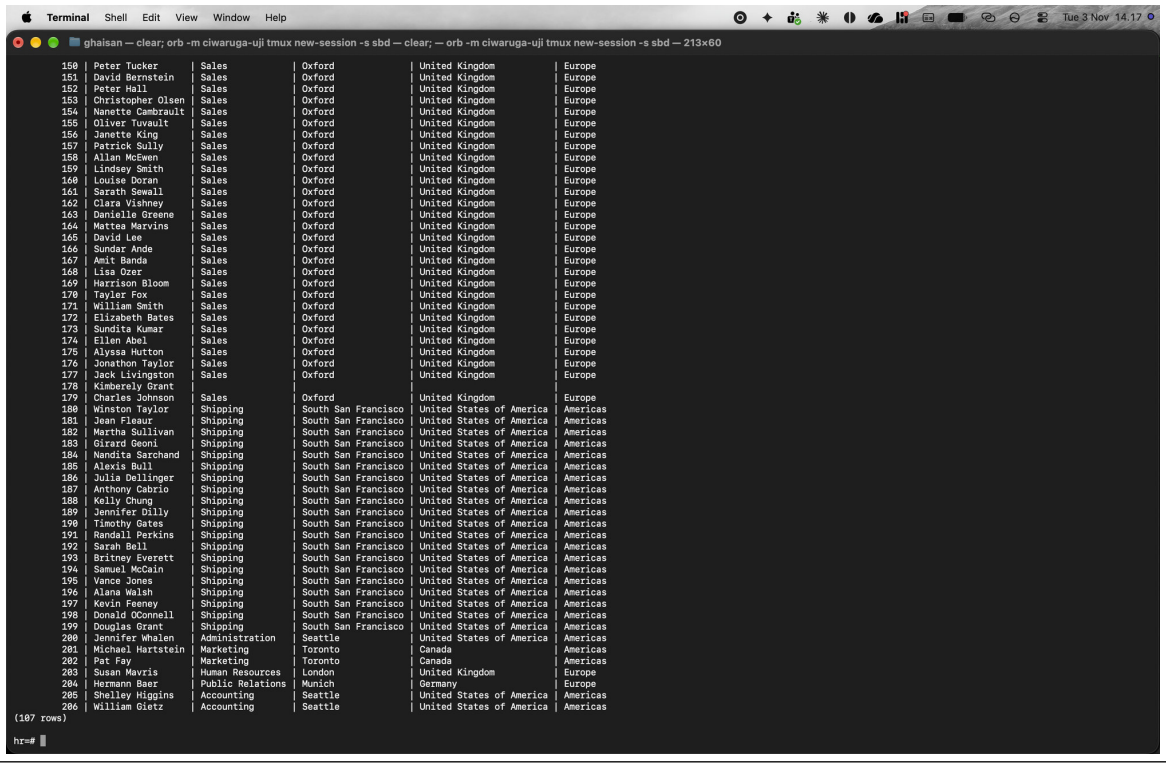
Keterangan

Data employee digabung berantai: employees ke departments (department_id), departments ke locations (location_id), locations ke countries (country_id), dan countries ke regions (region_id). Semua join memakai LEFT JOIN supaya employee yang tidak punya departemen tetap tampil. Kimberely Grant (ID 178) tidak punya department_id, sehingga kolom departemen, kota, negara, dan regionnya kosong. Hasilnya 107 baris, sama dengan jumlah employee.
Output lebih panjang dari satu layar, jadi ada dua screenshot: syntax dan awal output, lalu bagian akhir output.

Hasil Eksekusi (Screenshot)



```
Terminal  Shell  Edit  View  Window  Help                                                    Tue 3 Nov  14.17
ghaisan — clear; orb -m ciwaruga-uji tmux new-session -s sbd — clear; — orb -m ciwaruga-uji tmux new-session -s sbd — 213×60
hr=# SELECT e.employee_id, e.first_name || ' ' || e.last_name AS nama_employee,                                  14:17 [60/60]
hr-#        d.department_name, l.city, c.country_name, r.region_name
hr-# FROM employees e
hr-# LEFT JOIN departments d ON e.department_id = d.department_id
hr-# LEFT JOIN locations l ON d.location_id = l.location_id
hr-# LEFT JOIN countries c ON l.country_id = c.country_id
hr-# LEFT JOIN regions r ON c.region_id = r.region_id
hr-# ORDER BY e.employee_id;
 employee_id |   nama_employee   | department_name |       city        |       country_name       | region_name
-------------+-------------------+-----------------+-------------------+--------------------------+-------------
         100 | Steven King       | Executive       | Seattle           | United States of America | Americas
         101 | Neena Kochhar     | Executive       | Seattle           | United States of America | Americas
         102 | Lex De Haan       | Executive       | Seattle           | United States of America | Americas
         103 | Alexander Hunold  | IT              | Southlake         | United States of America | Americas
         104 | Bruce Ernst       | IT              | Southlake         | United States of America | Americas
         105 | David Austin      | IT              | Southlake         | United States of America | Americas
         106 | Valli Pataballa   | IT              | Southlake         | United States of America | Americas
         107 | Diana Lorentz     | IT              | Southlake         | United States of America | Americas
         108 | Nancy Greenberg   | Finance         | Seattle           | United States of America | Americas
         109 | Daniel Faviet     | Finance         | Seattle           | United States of America | Americas
         110 | John Chen         | Finance         | Seattle           | United States of America | Americas
         111 | Ismael Sciarra    | Finance         | Seattle           | United States of America | Americas
         112 | Jose Manuel Urman | Finance         | Seattle           | United States of America | Americas
         113 | Luis Popp         | Finance         | Seattle           | United States of America | Americas
         114 | Den Raphaely      | Purchasing      | Seattle           | United States of America | Americas
         115 | Alexander Khoo    | Purchasing      | Seattle           | United States of America | Americas
         116 | Shelli Baida      | Purchasing      | Seattle           | United States of America | Americas
         117 | Sigal Tobias      | Purchasing      | Seattle           | United States of America | Americas
         118 | Guy Himuro        | Purchasing      | Seattle           | United States of America | Americas
         119 | Karen Colmenares  | Purchasing      | Seattle           | United States of America | Americas
         120 | Matthew Weiss     | Shipping        | South San Francisco | United States of America | Americas
         121 | Adam Fripp        | Shipping        | South San Francisco | United States of America | Americas
         122 | Payam Kaufling    | Shipping        | South San Francisco | United States of America | Americas
         123 | Shanta Vollman    | Shipping        | South San Francisco | United States of America | Americas
         124 | Kevin Mourgos     | Shipping        | South San Francisco | United States of America | Americas
         125 | Julia Nayer       | Shipping        | South San Francisco | United States of America | Americas
         126 | Irene Mikkilineni | Shipping        | South San Francisco | United States of America | Americas
         127 | James Landry      | Shipping        | South San Francisco | United States of America | Americas
         128 | Steven Markle     | Shipping        | South San Francisco | United States of America | Americas
         129 | Laura Bissot      | Shipping        | South San Francisco | United States of America | Americas
         130 | Mozhe Atkinson    | Shipping        | South San Francisco | United States of America | Americas
         131 | James Marlow      | Shipping        | South San Francisco | United States of America | Americas
         132 | TJ Olson          | Shipping        | South San Francisco | United States of America | Americas
         133 | Jason Mallin      | Shipping        | South San Francisco | United States of America | Americas
         134 | Michael Rogers    | Shipping        | South San Francisco | United States of America | Americas
         135 | Ki Gee            | Shipping        | South San Francisco | United States of America | Americas
         136 | Hazel Philtanker  | Shipping        | South San Francisco | United States of America | Americas
         137 | Renske Ladwig     | Shipping        | South San Francisco | United States of America | Americas
         138 | Stephen Stiles    | Shipping        | South San Francisco | United States of America | Americas
         139 | John Seo          | Shipping        | South San Francisco | United States of America | Americas
         140 | Joshua Patel      | Shipping        | South San Francisco | United States of America | Americas
         141 | Trenna Rajs       | Shipping        | South San Francisco | United States of America | Americas
         142 | Curtis Davies     | Shipping        | South San Francisco | United States of America | Americas
         143 | Randall Matos     | Shipping        | South San Francisco | United States of America | Americas
         144 | Peter Vargas      | Shipping        | South San Francisco | United States of America | Americas
         145 | John Russell      | Sales           | Oxford            | United Kingdom           | Europe
         146 | Karen Partners    | Sales           | Oxford            | United Kingdom           | Europe
         147 | Alberto Errazuriz | Sales           | Oxford            | United Kingdom           | Europe
         148 | Gerald Cambrault  | Sales           | Oxford            | United Kingdom           | Europe
         149 | Eleni Zlotkey     | Sales           | Oxford            | United Kingdom           | Europe
```



```
Terminal  Shell  Edit  View  Window  Help                                                    Tue 3 Nov  14.17
ghaisan — clear; orb -m ciwaruga-uji tmux new-session -s sbd — clear; — orb -m ciwaruga-uji tmux new-session -s sbd — 213×60
         150 | Peter Tucker      | Sales           | Oxford            | United Kingdom           | Europe
         151 | David Bernstein   | Sales           | Oxford            | United Kingdom           | Europe
         152 | Peter Hall        | Sales           | Oxford            | United Kingdom           | Europe
         153 | Christopher Olsen | Sales           | Oxford            | United Kingdom           | Europe
         154 | Nanette Cambrault | Sales           | Oxford            | United Kingdom           | Europe
         155 | Oliver Tuvault    | Sales           | Oxford            | United Kingdom           | Europe
         156 | Janette King      | Sales           | Oxford            | United Kingdom           | Europe
         157 | Patrick Sully     | Sales           | Oxford            | United Kingdom           | Europe
         158 | Allan McEwen      | Sales           | Oxford            | United Kingdom           | Europe
         159 | Lindsey Smith     | Sales           | Oxford            | United Kingdom           | Europe
         160 | Louise Doran      | Sales           | Oxford            | United Kingdom           | Europe
         161 | Sarath Sewall     | Sales           | Oxford            | United Kingdom           | Europe
         162 | Clara Vishney     | Sales           | Oxford            | United Kingdom           | Europe
         163 | Danielle Greene   | Sales           | Oxford            | United Kingdom           | Europe
         164 | Mattea Marvins    | Sales           | Oxford            | United Kingdom           | Europe
         165 | David Lee         | Sales           | Oxford            | United Kingdom           | Europe
         166 | Sundar Ande       | Sales           | Oxford            | United Kingdom           | Europe
         167 | Amit Banda        | Sales           | Oxford            | United Kingdom           | Europe
         168 | Lisa Ozer         | Sales           | Oxford            | United Kingdom           | Europe
         169 | Harrison Bloom    | Sales           | Oxford            | United Kingdom           | Europe
         170 | Tayler Fox        | Sales           | Oxford            | United Kingdom           | Europe
         171 | William Smith     | Sales           | Oxford            | United Kingdom           | Europe
         172 | Elizabeth Bates   | Sales           | Oxford            | United Kingdom           | Europe
         173 | Sundita Kumar     | Sales           | Oxford            | United Kingdom           | Europe
         174 | Ellen Abel        | Sales           | Oxford            | United Kingdom           | Europe
         175 | Alyssa Hutton     | Sales           | Oxford            | United Kingdom           | Europe
         176 | Jonathon Taylor   | Sales           | Oxford            | United Kingdom           | Europe
         177 | Jack Livingston   | Sales           | Oxford            | United Kingdom           | Europe
         178 | Kimberely Grant   |                 |                   |                          |
         179 | Charles Johnson   | Sales           | Oxford            | United Kingdom           | Europe
         180 | Winston Taylor    | Shipping        | South San Francisco | United States of America | Americas
         181 | Jean Fleaur       | Shipping        | South San Francisco | United States of America | Americas
         182 | Martha Sullivan   | Shipping        | South San Francisco | United States of America | Americas
         183 | Girard Geoni      | Shipping        | South San Francisco | United States of America | Americas
         184 | Nandita Sarchand  | Shipping        | South San Francisco | United States of America | Americas
         185 | Alexis Bull       | Shipping        | South San Francisco | United States of America | Americas
         186 | Julia Dellinger   | Shipping        | South San Francisco | United States of America | Americas
         187 | Anthony Cabrio    | Shipping        | South San Francisco | United States of America | Americas
         188 | Kelly Chung       | Shipping        | South San Francisco | United States of America | Americas
         189 | Jennifer Dilly    | Shipping        | South San Francisco | United States of America | Americas
         190 | Timothy Gates     | Shipping        | South San Francisco | United States of America | Americas
         191 | Randall Perkins   | Shipping        | South San Francisco | United States of America | Americas
         192 | Sarah Bell        | Shipping        | South San Francisco | United States of America | Americas
         193 | Britney Everett   | Shipping        | South San Francisco | United States of America | Americas
         194 | Samuel McCain     | Shipping        | South San Francisco | United States of America | Americas
         195 | Vance Jones       | Shipping        | South San Francisco | United States of America | Americas
         196 | Alana Walsh       | Shipping        | South San Francisco | United States of America | Americas
         197 | Kevin Feeney      | Shipping        | South San Francisco | United States of America | Americas
         198 | Donald OConnell   | Shipping        | South San Francisco | United States of America | Americas
         199 | Douglas Grant     | Shipping        | South San Francisco | United States of America | Americas
         200 | Jennifer Whalen   | Administration  | Seattle           | United States of America | Americas
         201 | Michael Hartstein | Marketing       | Toronto           | Canada                   | Americas
         202 | Pat Fay           | Marketing       | Toronto           | Canada                   | Americas
         203 | Susan Mavris      | Human Resources | London            | United Kingdom           | Europe
         204 | Hermann Baer      | Public Relations | Munich           | Germany                  | Europe
         205 | Shelley Higgins   | Accounting      | Seattle           | United States of America | Americas
         206 | William Gietz     | Accounting      | Seattle           | United States of America | Americas
(107 rows)

hr=#
```

Latihan No. 3

Soal

Tampilkan Total Pendapatan Employee yang diperoleh dari Salary dan Commission Percentage!

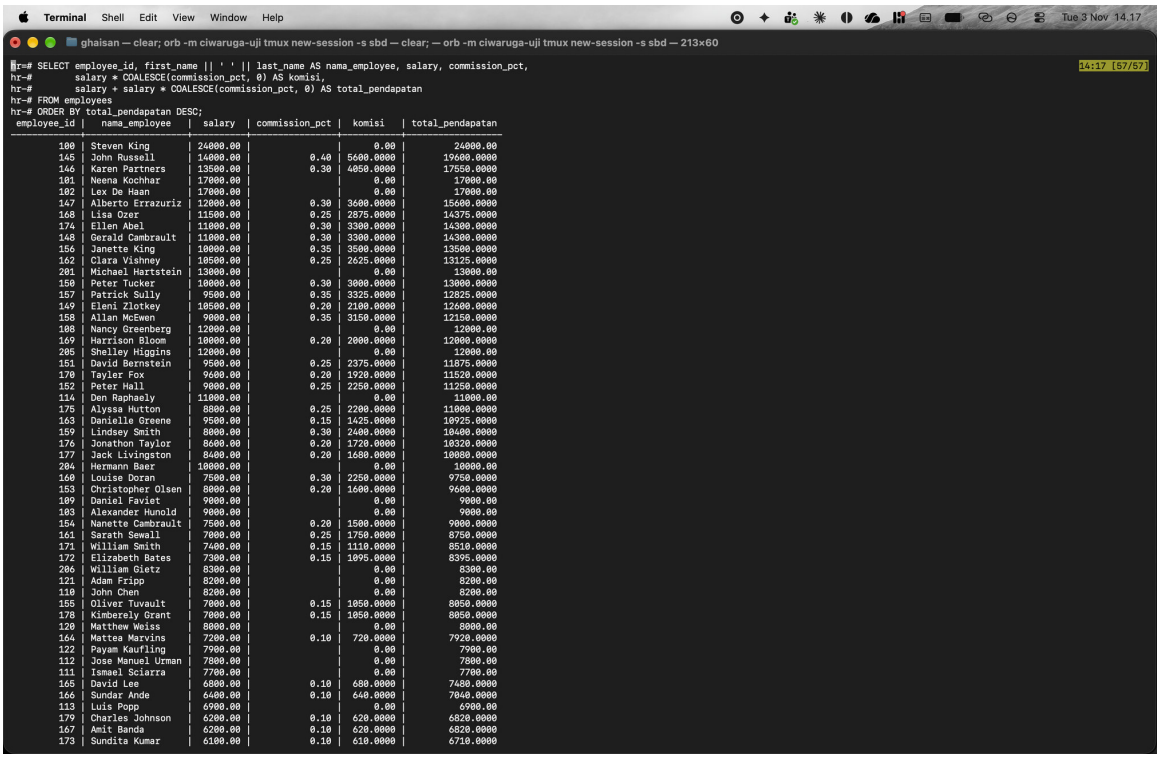
Syntax

```
SELECT employee_id, first_name || ' ' || last_name AS nama_employee, salary,
    commission_pct,
       salary * COALESCE(commission_pct, 0) AS komisi,
       salary + salary * COALESCE(commission_pct, 0) AS total_pendapatan
FROM employees
ORDER BY total_pendapatan DESC;
```

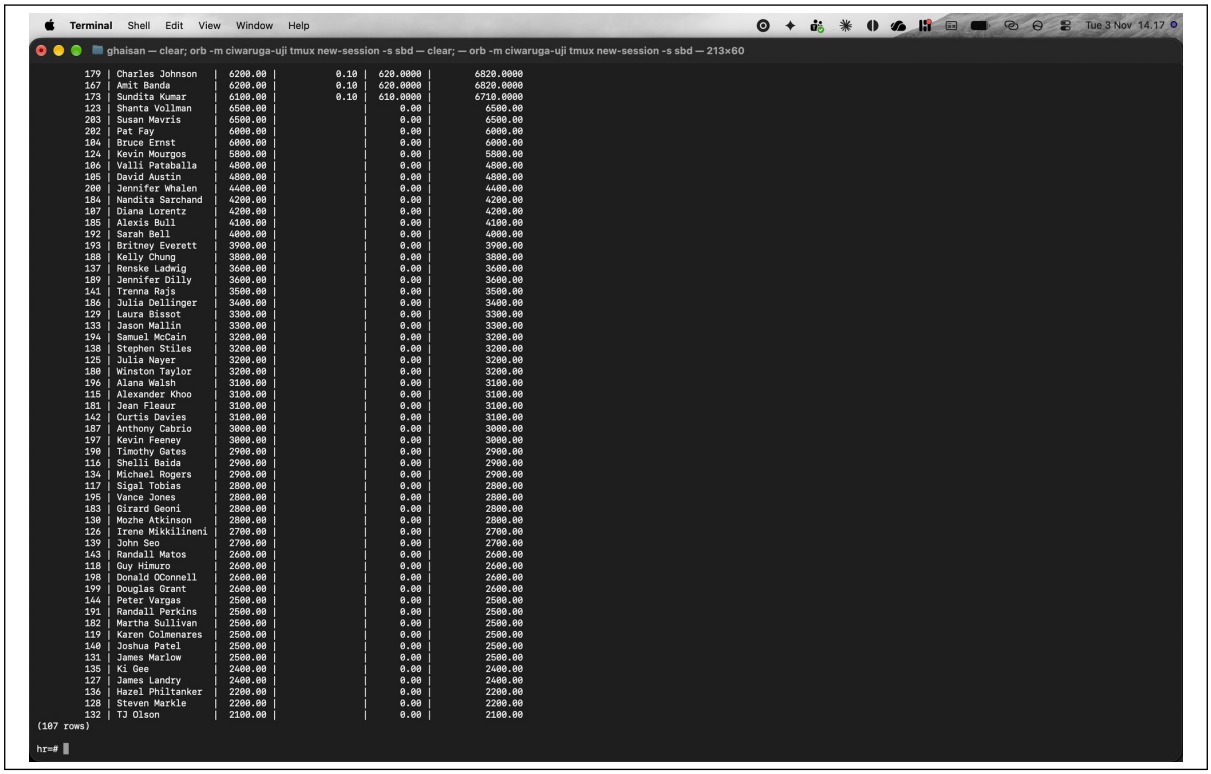
Keterangan

commission_pct berisi persentase komisi dalam bentuk pecahan (misalnya 0.40 = 40%), jadi komisi = salary x commission_pct dan total pendapatan = salary + komisi.
Employee yang bukan sales tidak punya commission_pct (NULL). Tanpa COALESCE, salary + NULL hasilnya NULL, jadi COALESCE(commission_pct, 0) dipakai supaya komisinya 0 dan total pendapatannya sama dengan salary.
Hasil diurutkan dari total pendapatan terbesar: Steven King 24000, lalu John Russell 14000 + 5600 = 19600. Ada 107 baris.
Output lebih panjang dari satu layar, jadi ada dua screenshot: syntax dan awal output, lalu bagian akhir output.

Hasil Eksekusi (Screenshot)



```
Terminal  Shell  Edit  View  Window  Help                                                    Tue 3 Nov  14.17
ghaisan — clear; orb -m ciwaruga-uji tmux new-session -s sbd — clear; — orb -m ciwaruga-uji tmux new-session -s sbd — 213×60
hr=# SELECT employee_id, first_name || ' ' || last_name AS nama_employee, salary, commission_pct,                14:17 [57/57]
hr-#        salary * COALESCE(commission_pct, 0) AS komisi,
hr-#        salary + salary * COALESCE(commission_pct, 0) AS total_pendapatan
hr-# FROM employees
hr-# ORDER BY total_pendapatan DESC;
 employee_id |    nama_employee    |  salary  | commission_pct |  komisi   | total_pendapatan
-------------+---------------------+----------+----------------+-----------+------------------
         100 | Steven King         | 24000.00 |                |      0.00 |         24000.00
         145 | John Russell        | 14000.00 |           0.40 | 5600.0000 |         19600.00
         146 | Karen Partners      | 13500.00 |           0.30 | 4050.0000 |         17550.00
         101 | Neena Kochhar       | 17000.00 |                |      0.00 |         17000.00
         102 | Lex De Haan         | 17000.00 |                |      0.00 |         17000.00
         147 | Alberto Errazuriz   | 12000.00 |           0.30 | 3600.0000 |         15600.00
         148 | Lisa Ozer           | 11500.00 |           0.25 | 2875.0000 |         14375.00
         174 | Ellen Abel          | 11000.00 |           0.30 | 3300.0000 |         14300.00
         148 | Gerald Cambrault    | 11000.00 |           0.30 | 3300.0000 |         14300.00
         156 | Janette King        | 10000.00 |           0.35 | 3500.0000 |         13500.00
         162 | Clara Vishney       | 10500.00 |           0.25 | 2625.0000 |         13125.00
         201 | Michael Hartstein   | 13000.00 |                |      0.00 |         13000.00
         150 | Peter Tucker        | 10000.00 |           0.30 | 3000.0000 |         13000.00
         157 | Patrick Sully       |  9500.00 |           0.35 | 3325.0000 |         12825.00
         149 | Eleni Zlotkey       | 10500.00 |           0.20 | 2100.0000 |         12600.00
         158 | Allan McEwen        |  9000.00 |           0.35 | 3150.0000 |         12150.00
         108 | Nancy Greenberg     | 12000.00 |                |      0.00 |         12000.00
         169 | Harrison Bloom      | 10000.00 |           0.20 | 2000.0000 |         12000.00
         205 | Shelley Higgins     | 12000.00 |                |      0.00 |         12000.00
         151 | David Bernstein     |  9500.00 |           0.25 | 2375.0000 |         11875.00
         170 | Tayler Fox          |  9600.00 |           0.20 | 1920.0000 |         11520.00
         152 | Peter Hall          |  9000.00 |           0.25 | 2250.0000 |         11250.00
         114 | Den Raphaely        | 11000.00 |                |      0.00 |         11000.00
         175 | Alyssa Hutton       |  8800.00 |           0.25 | 2200.0000 |         11000.00
         163 | Danielle Greene     |  9500.00 |           0.15 | 1425.0000 |         10925.00
         159 | Lindsey Smith       |  8000.00 |           0.30 | 2400.0000 |         10400.00
         176 | Jonathon Taylor     |  8600.00 |           0.20 | 1720.0000 |         10320.00
         177 | Jack Livingston     |  8400.00 |           0.20 | 1680.0000 |         10080.00
         204 | Hermann Baer        | 10000.00 |                |      0.00 |         10000.00
         160 | Louise Doran        |  7500.00 |           0.30 | 2250.0000 |          9750.00
         153 | Christopher Olsen   |  8000.00 |           0.20 | 1600.0000 |          9600.00
         109 | Daniel Faviet       |  9000.00 |                |      0.00 |          9000.00
         103 | Alexander Hunold    |  9000.00 |                |      0.00 |          9000.00
         154 | Nanette Cambrault   |  7500.00 |           0.20 | 1500.0000 |          9000.00
         161 | Sarath Sewall       |  7000.00 |           0.25 | 1750.0000 |          8750.00
         171 | William Smith       |  7400.00 |           0.15 | 1110.0000 |          8510.00
         172 | Elizabeth Bates     |  7300.00 |           0.15 | 1095.0000 |          8395.00
         206 | William Gietz       |  8300.00 |                |      0.00 |          8300.00
         121 | Adam Fripp          |  8200.00 |                |      0.00 |          8200.00
         110 | John Chen           |  8200.00 |                |      0.00 |          8200.00
         155 | Oliver Tuvault      |  7000.00 |           0.15 | 1050.0000 |          8050.00
         178 | Kimberely Grant     |  7000.00 |           0.15 | 1050.0000 |          8050.00
         120 | Matthew Weiss       |  8000.00 |                |      0.00 |          8000.00
         164 | Mattea Marvins      |  7200.00 |           0.10 |  720.0000 |          7920.00
         122 | Payam Kaufling      |  7900.00 |                |      0.00 |          7900.00
         112 | Jose Manuel Urman   |  7800.00 |                |      0.00 |          7800.00
         111 | Ismael Sciarra      |  7700.00 |                |      0.00 |          7700.00
         165 | David Lee           |  6800.00 |           0.10 |  680.0000 |          7480.00
         166 | Sundar Ande         |  6400.00 |           0.10 |  640.0000 |          7040.00
         113 | Luis Popp           |  6900.00 |                |      0.00 |          6900.00
         179 | Charles Johnson     |  6200.00 |           0.10 |  620.0000 |          6820.00
         167 | Amit Banda          |  6200.00 |           0.10 |  620.0000 |          6820.00
         173 | Sundita Kumar       |  6100.00 |           0.10 |  610.0000 |          6710.00
```



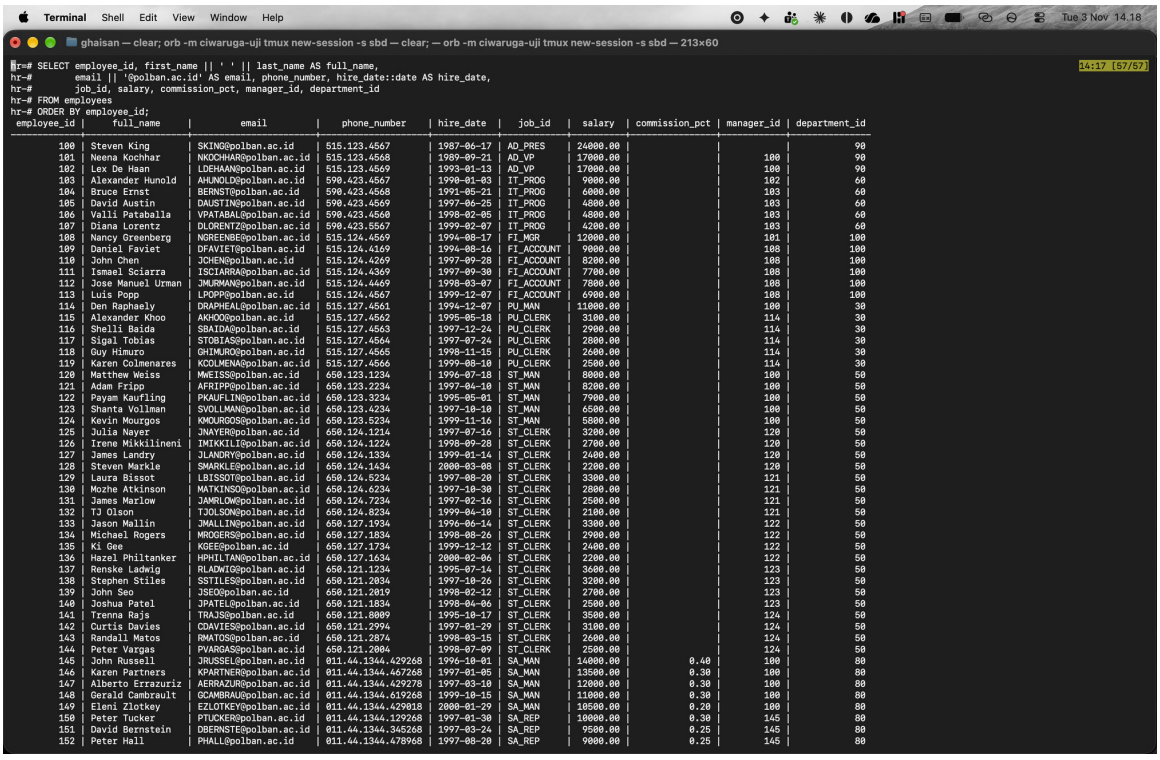
Latihan No. 4

Soal
Tampilkan seluruh Data Employee dengan kolom yang digabung antara First Name dan Last Name, dan tambahan string pada email @polban.ac.id!
Syntax

`SELECT employee_id, first_name || ' ' || last_name AS full_name,`
`email || '@polban.ac.id' AS email, phone_number, hire_date::date AS hire_date,`
`job_id, salary, commission_pct, manager_id, department_id`
`FROM employees`
`ORDER BY employee_id;`

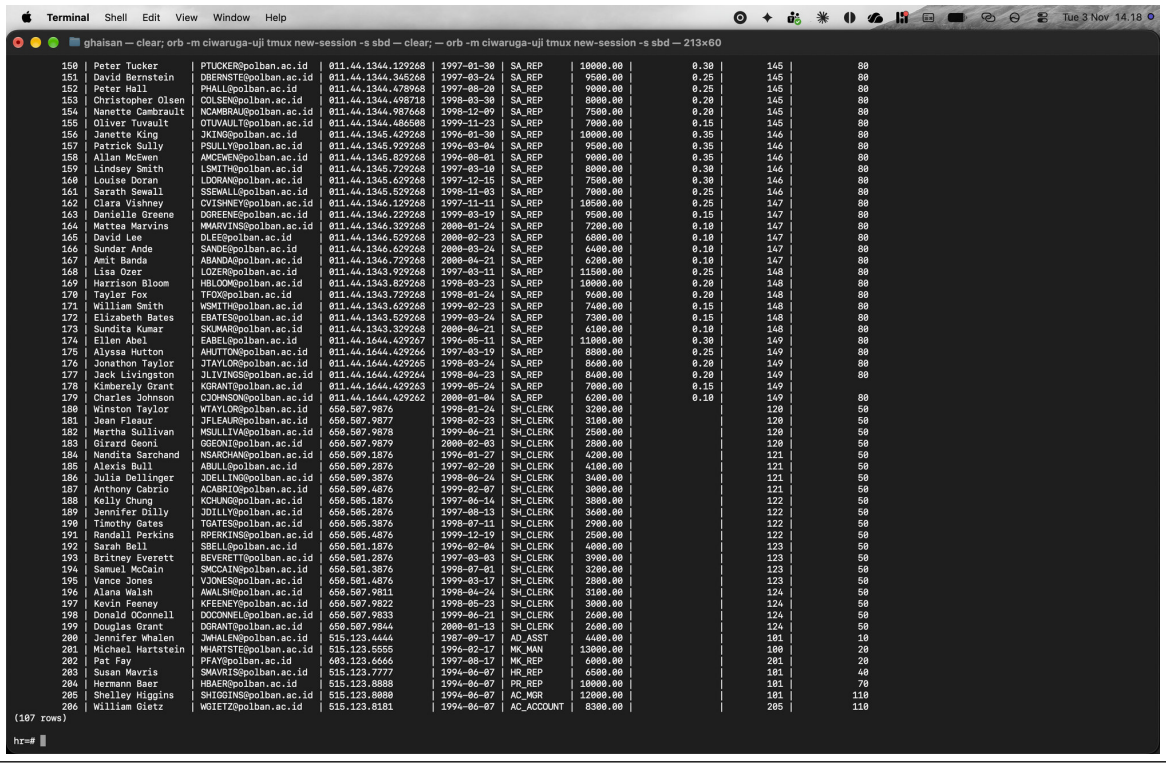
Keterangan
first_name dan last_name digabung dengan operator || dan spasi di tengahnya menjadi kolom full_name. Kolom email ditambah string '@polban.ac.id', misalnya SKING menjadi SKING@polban.ac.id. Kolom lain tetap ditampilkan, dan hire_date diubah ke tipe date supaya jamnya tidak ikut tampil. Hasilnya 107 baris (seluruh employee). Output lebih panjang dari satu layar, jadi ada dua screenshot: syntax dan awal output, lalu bagian akhir output.

Hasil Eksekusi (Screenshot)



```
Terminal  Shell  Edit  View  Window  Help                                                    Tue 3 Nov  14.18
ghaisan — clear; orb -m ciwaruga-uji tmux new-session -s sbd — clear; — orb -m ciwaruga-uji tmux new-session -s sbd — 213×60

hr=# SELECT employee_id, first_name || ' ' || last_name AS full_name,                                  14:17 [57/57]
hr-#        email || '@polban.ac.id' AS email, phone_number, hire_date::date AS hire_date,
hr-#        job_id, salary, commission_pct, manager_id, department_id
hr-# FROM employees
hr-# ORDER BY employee_id;
 employee_id |      full_name      |          email          |    phone_number    | hire_date |   job_id   | salary  | commission_pct | manager_id | department_id
-------------+---------------------+-------------------------+--------------------+-----------+------------+---------+----------------+------------+---------------
         100 | Steven King         | SKING@polban.ac.id      | 515.123.4567       | 1987-06-17 | AD_PRES    | 24000.00 |                |            |            90
         101 | Neena Kochhar       | NKOCHHAR@polban.ac.id   | 515.123.4568       | 1989-09-21 | AD_VP      | 17000.00 |                |        100 |            90
         102 | Lex De Haan         | LDEHAAN@polban.ac.id    | 515.123.4569       | 1993-01-13 | AD_VP      | 17000.00 |                |        100 |            90
         103 | Alexander Hunold    | AHUNOLD@polban.ac.id    | 590.423.4567       | 1990-01-03 | IT_PROG    |  9000.00 |                |        102 |            60
         104 | Bruce Ernst         | BERNST@polban.ac.id     | 590.423.4568       | 1991-05-21 | IT_PROG    |  6000.00 |                |        103 |            60
         105 | David Austin        | DAUSTIN@polban.ac.id    | 590.423.4569       | 1997-06-25 | IT_PROG    |  4800.00 |                |        103 |            60
         106 | Valli Pataballa     | VPATABAL@polban.ac.id   | 590.423.4560       | 1998-02-05 | IT_PROG    |  4800.00 |                |        103 |            60
         107 | Diana Lorentz       | DLORENTZ@polban.ac.id   | 590.423.5567       | 1999-02-07 | IT_PROG    |  4200.00 |                |        103 |            60
         108 | Nancy Greenberg     | NGREENBE@polban.ac.id   | 515.124.4569       | 1994-08-17 | FI_MGR     | 12000.00 |                |        101 |           100
         109 | Daniel Faviet       | DFAVIET@polban.ac.id    | 515.124.4169       | 1994-08-16 | FI_ACCOUNT |  9000.00 |                |        108 |           100
         110 | John Chen           | JCHEN@polban.ac.id      | 515.124.4269       | 1997-09-28 | FI_ACCOUNT |  8200.00 |                |        108 |           100
         111 | Ismael Sciarra      | ISCIARRA@polban.ac.id   | 515.124.4369       | 1997-09-30 | FI_ACCOUNT |  7700.00 |                |        108 |           100
         112 | Jose Manuel Urman   | JMURMAN@polban.ac.id    | 515.124.4469       | 1998-03-07 | FI_ACCOUNT |  7800.00 |                |        108 |           100
         113 | Luis Popp           | LPOPP@polban.ac.id      | 515.124.4567       | 1999-12-07 | FI_ACCOUNT |  6900.00 |                |        108 |           100
         114 | Den Raphaely        | DRAPHEAL@polban.ac.id   | 515.127.4561       | 1994-12-07 | PU_MAN     | 11000.00 |                |        100 |            30
         115 | Alexander Khoo      | AKHOO@polban.ac.id      | 515.127.4562       | 1995-05-18 | PU_CLERK   |  3100.00 |                |        114 |            30
         116 | Shelli Baida        | SBAIDA@polban.ac.id     | 515.127.4563       | 1997-12-24 | PU_CLERK   |  2900.00 |                |        114 |            30
         117 | Sigal Tobias        | STOBIAS@polban.ac.id    | 515.127.4564       | 1997-07-24 | PU_CLERK   |  2800.00 |                |        114 |            30
         118 | Guy Himuro          | GHIMURO@polban.ac.id    | 515.127.4565       | 1998-11-15 | PU_CLERK   |  2600.00 |                |        114 |            30
         119 | Karen Colmenares    | KCOLMENA@polban.ac.id   | 515.127.4566       | 1999-08-10 | PU_CLERK   |  2500.00 |                |        114 |            30
         120 | Matthew Weiss       | MWEISS@polban.ac.id     | 650.123.1234       | 1996-07-18 | ST_MAN     |  8000.00 |                |        100 |            50
         121 | Adam Fripp          | AFRIPP@polban.ac.id     | 650.123.2234       | 1997-04-10 | ST_MAN     |  8200.00 |                |        100 |            50
         122 | Payam Kaufling      | PKAUFLIN@polban.ac.id   | 650.123.3234       | 1995-05-01 | ST_MAN     |  7900.00 |                |        100 |            50
         123 | Shanta Vollman      | SVOLLMAN@polban.ac.id   | 650.123.4234       | 1997-10-10 | ST_MAN     |  6500.00 |                |        100 |            50
         124 | Kevin Mourgos       | KMOURGOS@polban.ac.id   | 650.123.5234       | 1999-11-16 | ST_MAN     |  5800.00 |                |        100 |            50
         125 | Julia Nayer         | JNAYER@polban.ac.id     | 650.124.1214       | 1997-07-16 | ST_CLERK   |  3200.00 |                |        120 |            50
         126 | Irene Mikkilineni   | IMIKKILI@polban.ac.id   | 650.124.1224       | 1998-09-28 | ST_CLERK   |  2700.00 |                |        120 |            50
         127 | James Landry        | JLANDRY@polban.ac.id    | 650.124.1334       | 1999-01-14 | ST_CLERK   |  2400.00 |                |        120 |            50
         128 | Steven Markle       | SMARKLE@polban.ac.id    | 650.124.1434       | 2000-03-08 | ST_CLERK   |  2200.00 |                |        120 |            50
         129 | Laura Bissot        | LBISSOT@polban.ac.id    | 650.124.5234       | 1997-08-20 | ST_CLERK   |  3300.00 |                |        121 |            50
         130 | Mozhe Atkinson      | MATKINSO@polban.ac.id   | 650.124.6234       | 1997-10-30 | ST_CLERK   |  2800.00 |                |        121 |            50
         131 | James Marlow        | JAMRLOW@polban.ac.id    | 650.124.7234       | 1997-02-16 | ST_CLERK   |  2500.00 |                |        121 |            50
         132 | TJ Olson            | TJOLSON@polban.ac.id    | 650.124.8234       | 1999-04-10 | ST_CLERK   |  2100.00 |                |        121 |            50
         133 | Jason Mallin        | JMALLIN@polban.ac.id    | 650.127.1934       | 1996-06-14 | ST_CLERK   |  3300.00 |                |        122 |            50
         134 | Michael Rogers      | MROGERS@polban.ac.id    | 650.127.1834       | 1998-08-26 | ST_CLERK   |  2900.00 |                |        122 |            50
         135 | Ki Gee              | KGEE@polban.ac.id       | 650.127.1734       | 1999-12-12 | ST_CLERK   |  2400.00 |                |        122 |            50
         136 | Hazel Philtanker    | HPHILTAN@polban.ac.id   | 650.127.1634       | 2000-02-06 | ST_CLERK   |  2200.00 |                |        122 |            50
         137 | Renske Ladwig       | RLADWIG@polban.ac.id    | 650.121.1234       | 1995-07-14 | ST_CLERK   |  3600.00 |                |        123 |            50
         138 | Stephen Stiles      | SSTILES@polban.ac.id    | 650.121.2034       | 1997-10-26 | ST_CLERK   |  3200.00 |                |        123 |            50
         139 | John Seo            | JSEO@polban.ac.id       | 650.121.2019       | 1998-02-12 | ST_CLERK   |  2700.00 |                |        123 |            50
         140 | Joshua Patel        | JPATEL@polban.ac.id     | 650.121.1834       | 1998-04-06 | ST_CLERK   |  2500.00 |                |        123 |            50
         141 | Trenna Rajs         | TRAJS@polban.ac.id      | 650.121.8009       | 1995-10-17 | ST_CLERK   |  3500.00 |                |        124 |            50
         142 | Curtis Davies       | CDAVIES@polban.ac.id    | 650.121.2994       | 1997-01-29 | ST_CLERK   |  3100.00 |                |        124 |            50
         143 | Randall Matos       | RMATOS@polban.ac.id     | 650.121.2874       | 1998-03-15 | ST_CLERK   |  2600.00 |                |        124 |            50
         144 | Peter Vargas        | PVARGAS@polban.ac.id    | 650.121.2004       | 1998-07-09 | ST_CLERK   |  2500.00 |                |        124 |            50
         145 | John Russell        | JRUSSEL@polban.ac.id    | 011.44.1344.429268 | 1996-10-01 | SA_MAN     | 14000.00 |           0.40 |        100 |            80
         146 | Karen Partners      | KPARTNER@polban.ac.id   | 011.44.1344.467268 | 1997-01-05 | SA_MAN     | 13500.00 |           0.30 |        100 |            80
         147 | Alberto Errazuriz   | AERRAZUR@polban.ac.id   | 011.44.1344.429278 | 1997-03-10 | SA_MAN     | 12000.00 |           0.30 |        100 |            80
         148 | Gerald Cambrault    | GCAMBRAU@polban.ac.id   | 011.44.1344.619268 | 1999-10-15 | SA_MAN     | 11000.00 |           0.30 |        100 |            80
         149 | Eleni Zlotkey       | EZLOTKEY@polban.ac.id   | 011.44.1344.429018 | 2000-01-29 | SA_MAN     | 10500.00 |           0.20 |        100 |            80
         150 | Peter Tucker        | PTUCKER@polban.ac.id    | 011.44.1344.129268 | 1997-01-30 | SA_REP     | 10000.00 |           0.30 |        145 |            80
         151 | David Bernstein     | DBERNSTE@polban.ac.id   | 011.44.1344.345268 | 1997-03-24 | SA_REP     |  9500.00 |           0.25 |        145 |            80
         152 | Peter Hall          | PHALL@polban.ac.id      | 011.44.1344.478968 | 1997-08-20 | SA_REP     |  9000.00 |           0.25 |        145 |            80
```



```
Terminal  Shell  Edit  View  Window  Help                                                    Tue 3 Nov  14.18
ghaisan — clear; orb -m ciwaruga-uji tmux new-session -s sbd — clear; — orb -m ciwaruga-uji tmux new-session -s sbd — 213×60

         150 | Peter Tucker        | PTUCKER@polban.ac.id    | 011.44.1344.129268 | 1997-01-30 | SA_REP     | 10000.00 |           0.30 |        145 |            80
         151 | David Bernstein     | DBERNSTE@polban.ac.id   | 011.44.1344.345268 | 1997-03-24 | SA_REP     |  9500.00 |           0.25 |        145 |            80
         152 | Peter Hall          | PHALL@polban.ac.id      | 011.44.1344.478968 | 1997-08-20 | SA_REP     |  9000.00 |           0.25 |        145 |            80
         153 | Christopher Olsen   | COLSEN@polban.ac.id     | 011.44.1344.498718 | 1998-03-30 | SA_REP     |  8000.00 |           0.20 |        145 |            80
         154 | Nanette Cambrault   | NCAMBRAU@polban.ac.id   | 011.44.1344.987668 | 1998-12-09 | SA_REP     |  7500.00 |           0.20 |        145 |            80
         155 | Oliver Tuvault      | OTUVAULT@polban.ac.id   | 011.44.1344.486508 | 1999-11-23 | SA_REP     |  7000.00 |           0.15 |        145 |            80
         156 | Janette King        | JKING@polban.ac.id      | 011.44.1345.429268 | 1996-01-30 | SA_REP     | 10000.00 |           0.35 |        146 |            80
         157 | Patrick Sully       | PSULLY@polban.ac.id     | 011.44.1345.929268 | 1996-03-04 | SA_REP     |  9500.00 |           0.35 |        146 |            80
         158 | Allan McEwen        | AMCEWEN@polban.ac.id    | 011.44.1345.829268 | 1996-08-01 | SA_REP     |  9000.00 |           0.35 |        146 |            80
         159 | Lindsey Smith       | LSMITH@polban.ac.id     | 011.44.1345.729268 | 1997-03-10 | SA_REP     |  8000.00 |           0.30 |        146 |            80
         160 | Louise Doran        | LDORAN@polban.ac.id     | 011.44.1345.629268 | 1997-12-15 | SA_REP     |  7500.00 |           0.30 |        146 |            80
         161 | Sarath Sewall       | SSEWALL@polban.ac.id    | 011.44.1345.529268 | 1998-11-03 | SA_REP     |  7000.00 |           0.25 |        146 |            80
         162 | Clara Vishney       | CVISHNEY@polban.ac.id   | 011.44.1346.129268 | 1997-11-11 | SA_REP     | 10500.00 |           0.25 |        147 |            80
         163 | Danielle Greene     | DGREENE@polban.ac.id    | 011.44.1346.229268 | 1999-03-19 | SA_REP     |  9500.00 |           0.15 |        147 |            80
         164 | Mattea Marvins      | MMARVINS@polban.ac.id   | 011.44.1346.329268 | 2000-01-24 | SA_REP     |  7200.00 |           0.10 |        147 |            80
         165 | David Lee           | DLEE@polban.ac.id       | 011.44.1346.529268 | 2000-02-23 | SA_REP     |  6800.00 |           0.10 |        147 |            80
         166 | Sundar Ande         | SANDE@polban.ac.id      | 011.44.1346.629268 | 2000-03-24 | SA_REP     |  6400.00 |           0.10 |        147 |            80
         167 | Amit Banda          | ABANDA@polban.ac.id     | 011.44.1346.729268 | 2000-04-21 | SA_REP     |  6200.00 |           0.10 |        147 |            80
         168 | Lisa Ozer           | LOZER@polban.ac.id      | 011.44.1343.929268 | 1997-03-11 | SA_REP     | 11500.00 |           0.25 |        148 |            80
         169 | Harrison Bloom      | HBLOOM@polban.ac.id     | 011.44.1343.829268 | 1998-03-23 | SA_REP     | 10000.00 |           0.20 |        148 |            80
         170 | Tayler Fox          | TFOX@polban.ac.id       | 011.44.1343.729268 | 1998-01-24 | SA_REP     |  9600.00 |           0.20 |        148 |            80
         171 | William Smith       | WSMITH@polban.ac.id     | 011.44.1343.629268 | 1999-02-23 | SA_REP     |  7400.00 |           0.15 |        148 |            80
         172 | Elizabeth Bates     | EBATES@polban.ac.id     | 011.44.1343.529268 | 1999-03-24 | SA_REP     |  7300.00 |           0.15 |        148 |            80
         173 | Sundita Kumar       | SKUMAR@polban.ac.id     | 011.44.1343.329268 | 2000-04-21 | SA_REP     |  6100.00 |           0.10 |        148 |            80
         174 | Ellen Abel          | EABEL@polban.ac.id      | 011.44.1644.429267 | 1996-05-11 | SA_REP     | 11000.00 |           0.30 |        149 |            80
         175 | Alyssa Hutton       | AHUTTON@polban.ac.id    | 011.44.1644.429266 | 1997-03-19 | SA_REP     |  8800.00 |           0.25 |        149 |            80
         176 | Jonathon Taylor     | JTAYLOR@polban.ac.id    | 011.44.1644.429265 | 1998-03-24 | SA_REP     |  8600.00 |           0.20 |        149 |            80
         177 | Jack Livingston     | JLIVINGS@polban.ac.id   | 011.44.1644.429264 | 1998-04-23 | SA_REP     |  8400.00 |           0.20 |        149 |            80
         178 | Kimberely Grant     | KGRANT@polban.ac.id     | 011.44.1644.429263 | 1999-05-24 | SA_REP     |  7000.00 |           0.15 |        149 |
         179 | Charles Johnson     | CJOHNSON@polban.ac.id   | 011.44.1644.429262 | 2000-01-04 | SA_REP     |  6200.00 |           0.10 |        149 |            80
         180 | Winston Taylor      | WTAYLOR@polban.ac.id    | 650.507.9876       | 1998-01-24 | SH_CLERK   |  3200.00 |                |        120 |            50
         181 | Jean Fleaur         | JFLEAUR@polban.ac.id    | 650.507.9877       | 1998-02-23 | SH_CLERK   |  3100.00 |                |        120 |            50
         182 | Martha Sullivan     | MSULLIVA@polban.ac.id   | 650.507.9878       | 1999-06-21 | SH_CLERK   |  2500.00 |                |        120 |            50
         183 | Girard Geoni        | GGEONI@polban.ac.id     | 650.507.9879       | 2000-02-03 | SH_CLERK   |  2800.00 |                |        120 |            50
         184 | Nandita Sarchand    | NSARCHAN@polban.ac.id   | 650.509.1876       | 1996-01-27 | SH_CLERK   |  4200.00 |                |        121 |            50
         185 | Alexis Bull         | ABULL@polban.ac.id      | 650.509.2876       | 1997-02-20 | SH_CLERK   |  4100.00 |                |        121 |            50
         186 | Julia Dellinger     | JDELLING@polban.ac.id   | 650.509.3876       | 1998-06-24 | SH_CLERK   |  3400.00 |                |        121 |            50
         187 | Anthony Cabrio      | ACABRIO@polban.ac.id    | 650.509.4876       | 1999-02-07 | SH_CLERK   |  3000.00 |                |        121 |            50
         188 | Kelly Chung         | KCHUNG@polban.ac.id     | 650.505.1876       | 1997-06-14 | SH_CLERK   |  3800.00 |                |        122 |            50
         189 | Jennifer Dilly      | JDILLY@polban.ac.id     | 650.505.2876       | 1997-08-13 | SH_CLERK   |  3600.00 |                |        122 |            50
         190 | Timothy Gates       | TGATES@polban.ac.id     | 650.505.3876       | 1998-07-11 | SH_CLERK   |  2900.00 |                |        122 |            50
         191 | Randall Perkins     | RPERKINS@polban.ac.id   | 650.505.4876       | 1999-12-19 | SH_CLERK   |  2500.00 |                |        122 |            50
         192 | Sarah Bell          | SBELL@polban.ac.id      | 650.501.1876       | 1996-02-04 | SH_CLERK   |  4000.00 |                |        123 |            50
         193 | Britney Everett     | BEVERETT@polban.ac.id   | 650.501.2876       | 1997-03-03 | SH_CLERK   |  3900.00 |                |        123 |            50
         194 | Samuel McCain       | SMCCAIN@polban.ac.id    | 650.501.3876       | 1998-07-01 | SH_CLERK   |  3200.00 |                |        123 |            50
         195 | Vance Jones         | VJONES@polban.ac.id     | 650.501.4876       | 1999-03-17 | SH_CLERK   |  2800.00 |                |        123 |            50
         196 | Alana Walsh         | AWALSH@polban.ac.id     | 650.507.9811       | 1998-04-24 | SH_CLERK   |  3100.00 |                |        124 |            50
         197 | Kevin Feeney        | KFEENEY@polban.ac.id    | 650.507.9822       | 1998-05-23 | SH_CLERK   |  3000.00 |                |        124 |            50
         198 | Donald OConnell     | DOCONNEL@polban.ac.id   | 650.507.9833       | 1999-06-21 | SH_CLERK   |  2600.00 |                |        124 |            50
         199 | Douglas Grant       | DGRANT@polban.ac.id     | 650.507.9844       | 2000-01-13 | SH_CLERK   |  2600.00 |                |        124 |            50
         200 | Jennifer Whalen     | JWHALEN@polban.ac.id    | 515.123.4444       | 1987-09-17 | AD_ASST    |  4400.00 |                |        101 |            10
         201 | Michael Hartstein   | MHARTSTE@polban.ac.id   | 515.123.5555       | 1996-02-17 | MK_MAN     | 13000.00 |                |        100 |            20
         202 | Pat Fay             | PFAY@polban.ac.id       | 603.123.6666       | 1997-08-17 | MK_REP     |  6000.00 |                |        201 |            20
         203 | Susan Mavris        | SMAVRIS@polban.ac.id    | 515.123.7777       | 1994-06-07 | HR_REP     |  6500.00 |                |        101 |            40
         204 | Hermann Baer        | HBAER@polban.ac.id      | 515.123.8888       | 1994-06-07 | PR_REP     | 10000.00 |                |        101 |            70
         205 | Shelley Higgins     | SHIGGINS@polban.ac.id   | 515.123.8080       | 1994-06-07 | AC_MGR     | 12000.00 |                |        101 |           110
         206 | William Gietz       | WGIETZ@polban.ac.id     | 515.123.8181       | 1994-06-07 | AC_ACCOUNT |  8300.00 |                |        205 |           110
(107 rows)

hr=#
```

Latihan No. 5

Soal

Tampilkan hanya Nama Employee, Nama Manager dari Employee tersebut yang memiliki masa kerja lebih dari 3 tahun!

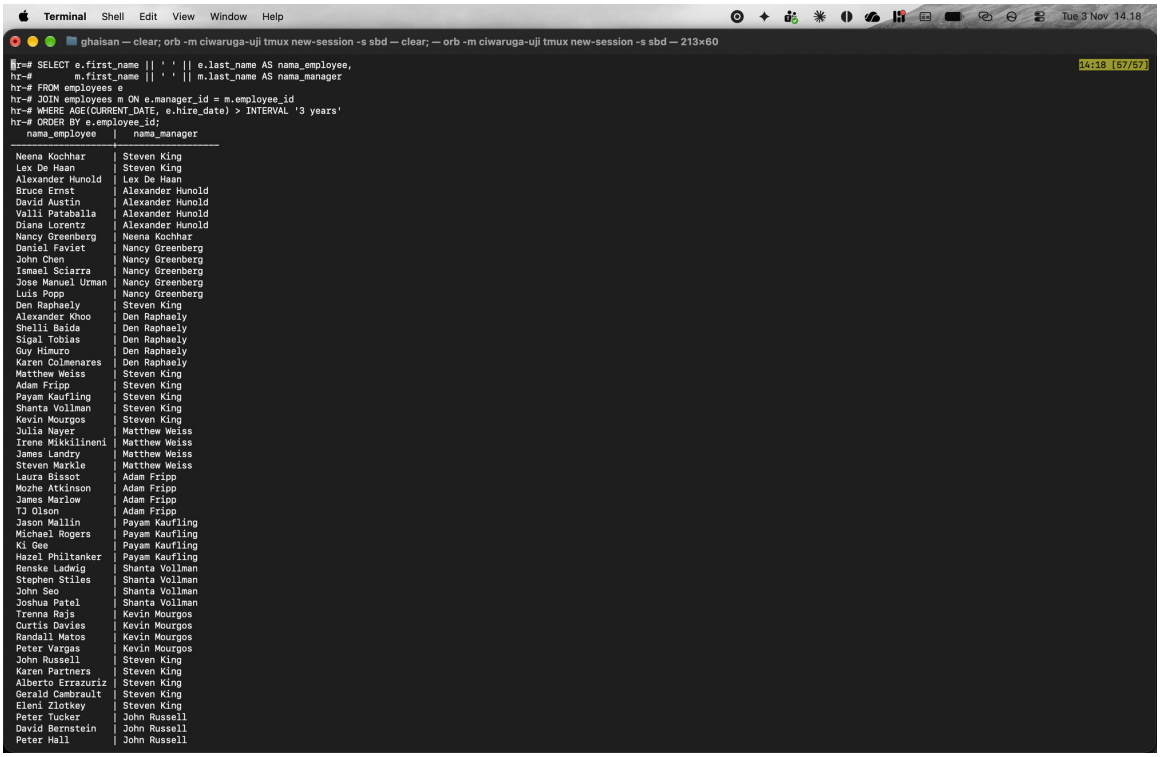
Syntax

```
SELECT e.first_name || ' ' || e.last_name AS nama_employee,
       m.first_name || ' ' || m.last_name AS nama_manager
FROM employees e
JOIN employees m ON e.manager_id = m.employee_id
WHERE AGE(CURRENT_DATE, e.hire_date) > INTERVAL '3 years'
ORDER BY e.employee_id;
```

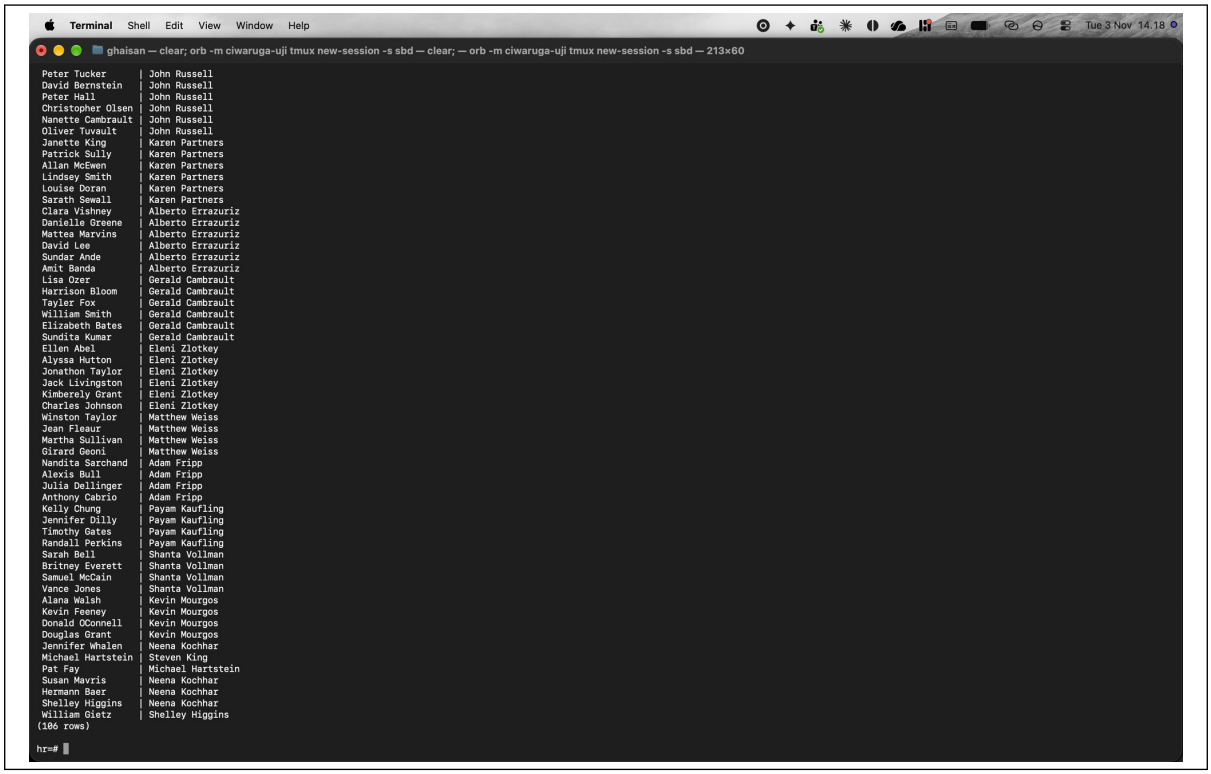
Keterangan

Nama manager diambil dengan self join: tabel employees dipakai dua kali, alias e untuk employee dan m untuk managernya, dihubungkan dengan e.manager_id = m.employee_id.
Masa kerja dihitung dengan AGE(CURRENT_DATE, e.hire_date) lalu dibandingkan dengan INTERVAL '3 years'.
Hasilnya 106 baris. Semua hire_date di schema HR ada di rentang 1987 sampai 2000, jadi semua employee masa kerjanya lebih dari 3 tahun. Satu employee tidak muncul, yaitu Steven King, karena tidak punya manager sehingga tidak ikut dalam inner join.
Output lebih panjang dari satu layar, jadi ada dua screenshot: syntax dan awal output, lalu bagian akhir output.

Hasil Eksekusi (Screenshot)



```
Terminal  Shell  Edit  View  Window  Help                                   Tue 3 Nov 14.18
ghaisan — clear; orb -m ciwaruga-uji tmux new-session -s sbd — clear; — orb -m ciwaruga-uji tmux new-session -s sbd — 213×60
hr-# SELECT e.first_name || ' ' || e.last_name AS nama_employee,                    14:18 [57/57]
hr-#        m.first_name || ' ' || m.last_name AS nama_manager
hr-# FROM employees e
hr-# JOIN employees m ON e.manager_id = m.employee_id
hr-# WHERE AGE(CURRENT_DATE, e.hire_date) > INTERVAL '3 years'
hr-# ORDER BY e.employee_id;
   nama_employee   |   nama_manager
-------------------+-------------------
 Neena Kochhar     | Steven King
 Lex De Haan       | Steven King
 Alexander Hunold  | Lex De Haan
 Bruce Ernst       | Alexander Hunold
 David Austin      | Alexander Hunold
 Valli Pataballa   | Alexander Hunold
 Diana Lorentz     | Alexander Hunold
 Nancy Greenberg   | Neena Kochhar
 Daniel Faviet     | Nancy Greenberg
 John Chen         | Nancy Greenberg
 Ismael Sciarra    | Nancy Greenberg
 Jose Manuel Urman | Nancy Greenberg
 Luis Popp         | Nancy Greenberg
 Den Raphaely      | Steven King
 Alexander Khoo    | Den Raphaely
 Shelli Baida      | Den Raphaely
 Sigal Tobias      | Den Raphaely
 Guy Himuro        | Den Raphaely
 Karen Colmenares  | Den Raphaely
 Matthew Weiss     | Steven King
 Adam Fripp        | Steven King
 Payam Kaufling    | Steven King
 Shanta Vollman    | Steven King
 Kevin Mourgos     | Steven King
 Julia Nayer       | Matthew Weiss
 Irene Mikkilineni | Matthew Weiss
 James Landry      | Matthew Weiss
 Steven Markle     | Matthew Weiss
 Laura Bissot      | Adam Fripp
 Mozhe Atkinson    | Adam Fripp
 James Marlow      | Adam Fripp
 TJ Olson          | Adam Fripp
 Jason Mallin      | Payam Kaufling
 Michael Rogers    | Payam Kaufling
 Ki Gee            | Payam Kaufling
 Hazel Philtanker  | Payam Kaufling
 Renske Ladwig     | Shanta Vollman
 Stephen Stiles    | Shanta Vollman
 John Seo          | Shanta Vollman
 Joshua Patel      | Shanta Vollman
 Trenna Rajs       | Kevin Mourgos
 Curtis Davies     | Kevin Mourgos
 Randall Matos     | Kevin Mourgos
 Peter Vargas      | Kevin Mourgos
 John Russell      | Steven King
 Karen Partners    | Steven King
 Alberto Errazuriz | Steven King
 Gerald Cambrault  | Steven King
 Eleni Zlotkey     | Steven King
 Peter Tucker      | John Russell
 David Bernstein   | John Russell
 Peter Hall        | John Russell
```



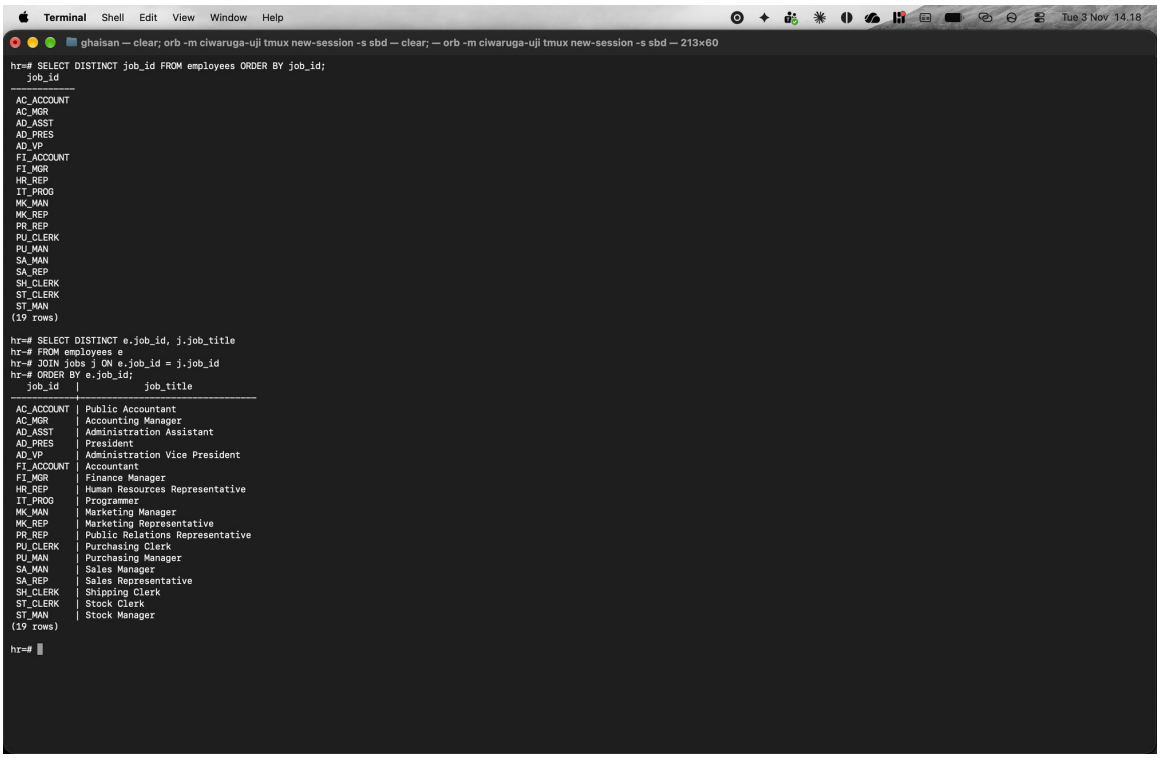
Latihan No. 6

Soal
Tampilkan seluruh jenis pekerjaan yang ada dari Tabel Employees tanpa duplikasi!
Syntax

`SELECT DISTINCT job_id FROM employees ORDER BY job_id;`
 	
`SELECT DISTINCT e.job_id, j.job_title`
`FROM employees e`
`JOIN jobs j ON e.job_id = j.job_id`
`ORDER BY e.job_id;`

Keterangan
DISTINCT menghilangkan job_id yang berulang, jadi setiap jenis pekerjaan hanya muncul sekali. Hasilnya 19 jenis pekerjaan dari 107 employee. Percobaan kedua menggabungkan tabel jobs supaya nama jabatannya (job_title) ikut tampil. Jumlahnya tetap 19 karena setiap job_id hanya punya satu job_title.

Hasil Eksekusi (Screenshot)



Latihan No. 7

Soal

Tampilkan Data Employee yang berada di Departement dengan ID: 50, namun tidak dipimpin oleh Manager dengan ID: 100!

Syntax

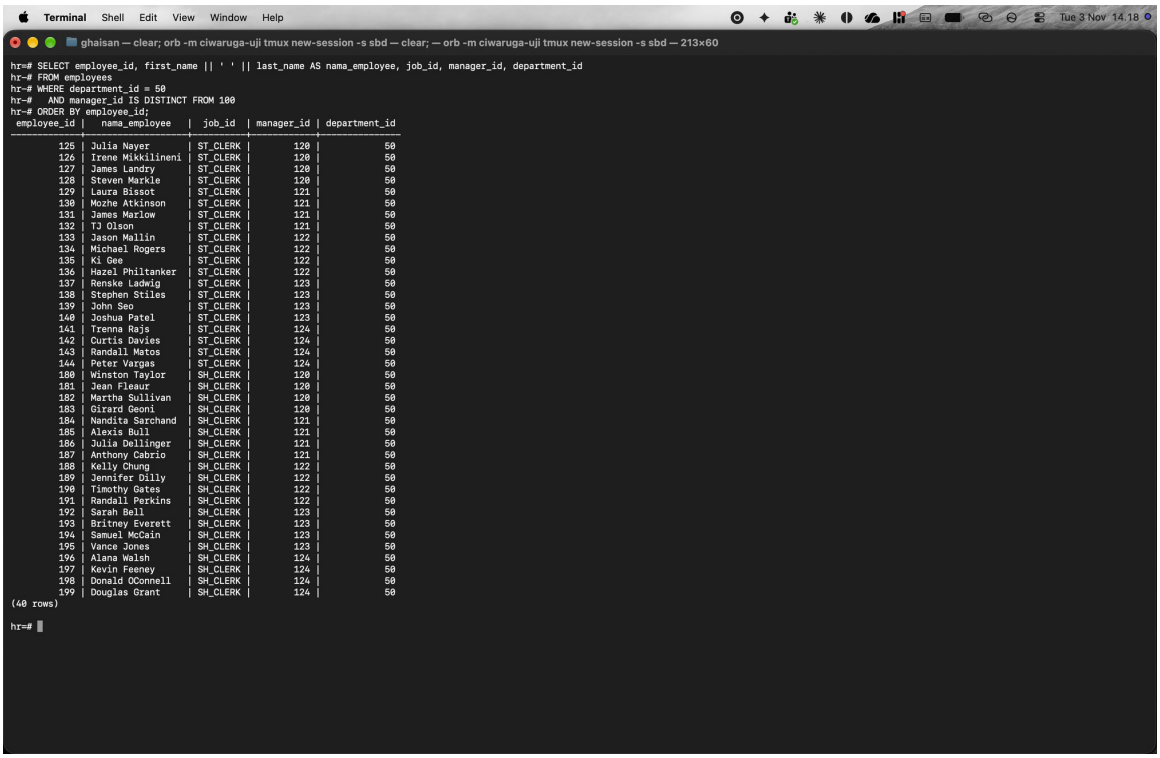
```
SELECT employee_id, first_name || ' ' || last_name AS nama_employee, job_id,
    manager_id, department_id
FROM employees
WHERE department_id = 50
  AND manager_id IS DISTINCT FROM 100
ORDER BY employee_id;
```

Keterangan

Department 50 adalah Shipping dengan 45 employee. Lima di antaranya (Stock Manager ID 120 sampai 124) dipimpin langsung oleh manager 100, sehingga disaring dengan manager_id IS DISTINCT FROM 100.

IS DISTINCT FROM dipakai, bukan <>, supaya employee yang manager_id-nya NULL tetap ikut tampil kalau ada (dengan <>, NULL tidak pernah memenuhi kondisi). Hasilnya 40 baris, semuanya Stock Clerk dan Shipping Clerk yang atasannya salah satu Stock Manager.

Hasil Eksekusi (Screenshot)



```
Terminal  Shell  Edit  View  Window  Help                                   Tue 3 Nov 14.18
ghaisan — clear; orb -m ciwaruga-uji tmux new-session -s sbd — clear; — orb -m ciwaruga-uji tmux new-session -s sbd — 213×60

hr=# SELECT employee_id, first_name || ' ' || last_name AS nama_employee, job_id, manager_id, department_id
hr-# FROM employees
hr-# WHERE department_id = 50
hr-#   AND manager_id IS DISTINCT FROM 100
hr-# ORDER BY employee_id;
 employee_id |   nama_employee   |  job_id  | manager_id | department_id
-------------+-------------------+----------+------------+---------------
         125 | Julia Nayer       | ST_CLERK |        120 |            50
         126 | Irene Mikkilineni | ST_CLERK |        120 |            50
         127 | James Landry      | ST_CLERK |        120 |            50
         128 | Steven Markle     | ST_CLERK |        120 |            50
         129 | Laura Bissot      | ST_CLERK |        121 |            50
         130 | Mozhe Atkinson    | ST_CLERK |        121 |            50
         131 | James Marlow      | ST_CLERK |        121 |            50
         132 | TJ Olson          | ST_CLERK |        121 |            50
         133 | Jason Mallin      | ST_CLERK |        122 |            50
         134 | Michael Rogers    | ST_CLERK |        122 |            50
         135 | Ki Gee            | ST_CLERK |        122 |            50
         136 | Hazel Philtanker  | ST_CLERK |        122 |            50
         137 | Renske Ladwig     | ST_CLERK |        123 |            50
         138 | Stephen Stiles    | ST_CLERK |        123 |            50
         139 | John Seo          | ST_CLERK |        123 |            50
         140 | Joshua Patel      | ST_CLERK |        123 |            50
         141 | Trenna Rajs       | ST_CLERK |        124 |            50
         142 | Curtis Davies     | ST_CLERK |        124 |            50
         143 | Randall Matos     | ST_CLERK |        124 |            50
         144 | Peter Vargas      | ST_CLERK |        124 |            50
         180 | Winston Taylor    | SH_CLERK |        120 |            50
         181 | Jean Fleaur       | SH_CLERK |        120 |            50
         182 | Martha Sullivan   | SH_CLERK |        120 |            50
         183 | Girard Geoni      | SH_CLERK |        120 |            50
         184 | Nandita Sarchand  | SH_CLERK |        121 |            50
         185 | Alexis Bull       | SH_CLERK |        121 |            50
         186 | Julia Dellinger   | SH_CLERK |        121 |            50
         187 | Anthony Cabrio    | SH_CLERK |        121 |            50
         188 | Kelly Chung       | SH_CLERK |        122 |            50
         189 | Jennifer Dilly    | SH_CLERK |        122 |            50
         190 | Timothy Gates     | SH_CLERK |        122 |            50
         191 | Randall Perkins   | SH_CLERK |        122 |            50
         192 | Sarah Bell        | SH_CLERK |        123 |            50
         193 | Britney Everett   | SH_CLERK |        123 |            50
         194 | Samuel McCain     | SH_CLERK |        123 |            50
         195 | Vance Jones       | SH_CLERK |        123 |            50
         196 | Alana Walsh       | SH_CLERK |        124 |            50
         197 | Kevin Feeney      | SH_CLERK |        124 |            50
         198 | Donald OConnell   | SH_CLERK |        124 |            50
         199 | Douglas Grant     | SH_CLERK |        124 |            50
(40 rows)

hr=#
```

Latihan No. 8

Soal

Bagian SDM ingin tahu seberapa lebar selisih gaji antara atasan dan bawahan langsungnya. Tampilkan hanya atasan yang memiliki minimal 5 bawahan langsung, dengan kolom:

- atasan (nama lengkap) dan job_title atasan
- departemen, diisi - jika atasan tidak punya departemen
- jumlah_bawahan
- rata_gaji_bawahan (2 desimal)
- rasio_persen, yaitu rata-rata gaji bawahan dibagi gaji atasan dikali 100 (1 desimal)
- kategori: rasio ≥ 75% = Selisih sempit, ≥ 40% = Normal, di bawah itu = Selisih lebar

Urutkan dari rasio terbesar, lalu nama atasan.

Syntax

```
SELECT m.first_name || ' ' || m.last_name AS atasan, j.job_title,
       COALESCE(d.department_name, '-') AS departemen,
       COUNT(b.employee_id) AS jumlah_bawahan,
       ROUND(AVG(b.salary), 2) AS rata_gaji_bawahan,
       ROUND(AVG(b.salary) / m.salary * 100, 1) AS rasio_persen,
       CASE
         WHEN AVG(b.salary) / m.salary * 100 >= 75 THEN 'Selisih sempit'
         WHEN AVG(b.salary) / m.salary * 100 >= 40 THEN 'Normal'
         ELSE 'Selisih lebar'
       END AS kategori
FROM employees m
JOIN employees b ON b.manager_id = m.employee_id
JOIN jobs j ON m.job_id = j.job_id
LEFT JOIN departments d ON m.department_id = d.department_id
GROUP BY m.employee_id, m.first_name, m.last_name, m.salary, j.job_title,
    d.department_name
```

```
HAVING COUNT(b.employee_id) >= 5
ORDER BY rasio_persen DESC, atasan;
```

Keterangan

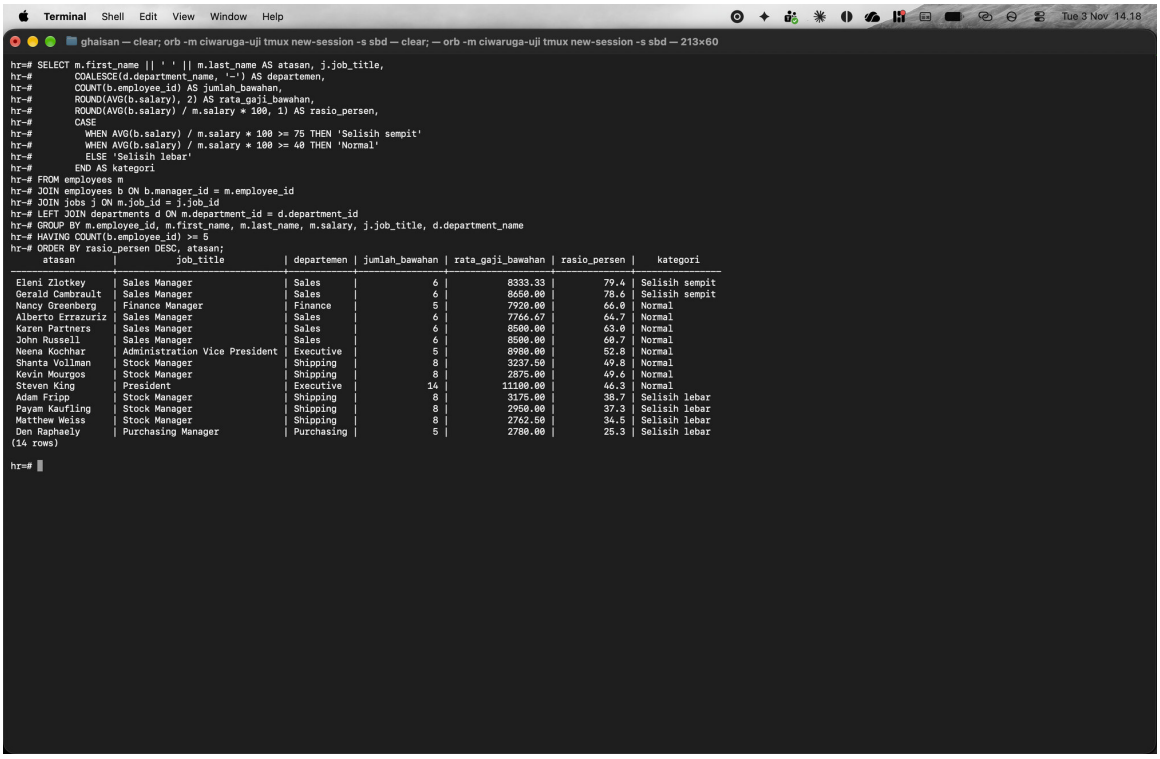
Tabel employees dipakai dua kali: m sebagai atasan dan b sebagai bawahan langsung (b.manager_id = m.employee_id). Data dikelompokkan per atasan, lalu HAVING COUNT(b.employee_id) >= 5 menyisakan atasan yang punya minimal 5 bawahan langsung.

job_title diambil dari tabel jobs. Departemen memakai LEFT JOIN dan COALESCE(d.department_name, '-') supaya atasan tanpa departemen tetap tampil dengan tanda -. Di data ini semua atasan punya departemen.

rata_gaji_bawahan = ROUND(AVG(b.salary), 2) dan rasio_persen = ROUND(AVG(b.salary) / m.salary x 100, 1). Kategori ditentukan dengan CASE dari rasio yang belum dibulatkan: minimal 75% Selisih sempit, minimal 40% Normal, sisanya Selisih lebar.

Hasilnya 14 atasan. Rasio terbesar Eleni Zlotkey 79,4% (Selisih sempit) dan terkecil Den Raphaely 25,3% (Selisih lebar). Steven King punya 14 bawahan langsung dengan rasio 46,3% (Normal). Tiga dari lima Stock Manager (Adam Fripp, Payam Kaufling, Matthew Weiss) masuk Selisih lebar.

Hasil Eksekusi (Screenshot)



```
Terminal  Shell  Edit  View  Window  Help                                   Tue 3 Nov 14.18
ghaisan — clear; orb -m ciwaruga-uji tmux new-session -s sbd — clear; — orb -m ciwaruga-uji tmux new-session -s sbd — 213×60

hr=# SELECT m.first_name || ' ' || m.last_name AS atasan, j.job_title,
hr-#        COALESCE(d.department_name, '-') AS departemen,
hr-#        COUNT(b.employee_id) AS jumlah_bawahan,
hr-#        ROUND(AVG(b.salary), 2) AS rata_gaji_bawahan,
hr-#        ROUND(AVG(b.salary) / m.salary * 100, 1) AS rasio_persen,
hr-#        CASE
hr-#          WHEN AVG(b.salary) / m.salary * 100 >= 75 THEN 'Selisih sempit'
hr-#          WHEN AVG(b.salary) / m.salary * 100 >= 40 THEN 'Normal'
hr-#          ELSE 'Selisih lebar'
hr-#        END AS kategori
hr-# FROM employees m
hr-# JOIN employees b ON b.manager_id = m.employee_id
hr-# JOIN jobs j ON m.job_id = j.job_id
hr-# LEFT JOIN departments d ON m.department_id = d.department_id
hr-# GROUP BY m.employee_id, m.first_name, m.last_name, m.salary, j.job_title, d.department_name
hr-# HAVING COUNT(b.employee_id) >= 5
hr-# ORDER BY rasio_persen DESC, atasan;
     atasan      |          job_title          | departemen | jumlah_bawahan | rata_gaji_bawahan | rasio_persen |    kategori
-----------------+-----------------------------+------------+----------------+-------------------+--------------+----------------
 Eleni Zlotkey   | Sales Manager               | Sales      |              6 |           8333.33 |         79.4 | Selisih sempit
 Gerald Cambrault | Sales Manager              | Sales      |              6 |           8650.00 |         78.6 | Selisih sempit
 Nancy Greenberg | Finance Manager             | Finance    |              5 |           7920.00 |         66.0 | Normal
 Alberto Errazuriz | Sales Manager             | Sales      |              6 |           7766.67 |         64.7 | Normal
 Karen Partners  | Sales Manager               | Sales      |              6 |           8500.00 |         63.0 | Normal
 John Russell    | Sales Manager               | Sales      |              6 |           8500.00 |         60.7 | Normal
 Neena Kochhar   | Administration Vice President | Executive |              5 |           8980.00 |         52.8 | Normal
 Shanta Vollman  | Stock Manager               | Shipping   |              8 |           3237.50 |         49.8 | Normal
 Kevin Mourgos   | Stock Manager               | Shipping   |              8 |           2875.00 |         49.6 | Normal
 Steven King     | President                   | Executive  |             14 |          11100.00 |         46.3 | Normal
 Adam Fripp      | Stock Manager               | Shipping   |              8 |           3175.00 |         38.7 | Selisih lebar
 Payam Kaufling  | Stock Manager               | Shipping   |              8 |           2950.00 |         37.3 | Selisih lebar
 Matthew Weiss   | Stock Manager               | Shipping   |              8 |           2762.50 |         34.5 | Selisih lebar
 Den Raphaely    | Purchasing Manager          | Purchasing |              5 |           2780.00 |         25.3 | Selisih lebar
(14 rows)

hr=#
```

Latihan No. 9

Soal

Sebelum migrasi data, auditor meminta satu laporan berisi semua anomali.
Buat satu query dengan tiga kolom (jenis_temuan, kode, keterangan) yang memuat empat jenis temuan:

1. PEGAWAI TANPA DEPARTEMEN: kode = employee_id, keterangan = nama lengkap.
2. DEPARTEMEN TANPA PEGAWAI: kode = department_id, keterangan = nama departemen.
3. LOKASI TIDAK DIPAKAI DEPARTEMEN: kode = location_id, keterangan = kota, negara.
4. GAJI LEBIH BESAR DARI ATASAN: kode = employee_id, keterangan = Nama (gaji) > Nama Atasan (gaji atasan).

Urutkan menurut jenis_temuan, lalu kode.

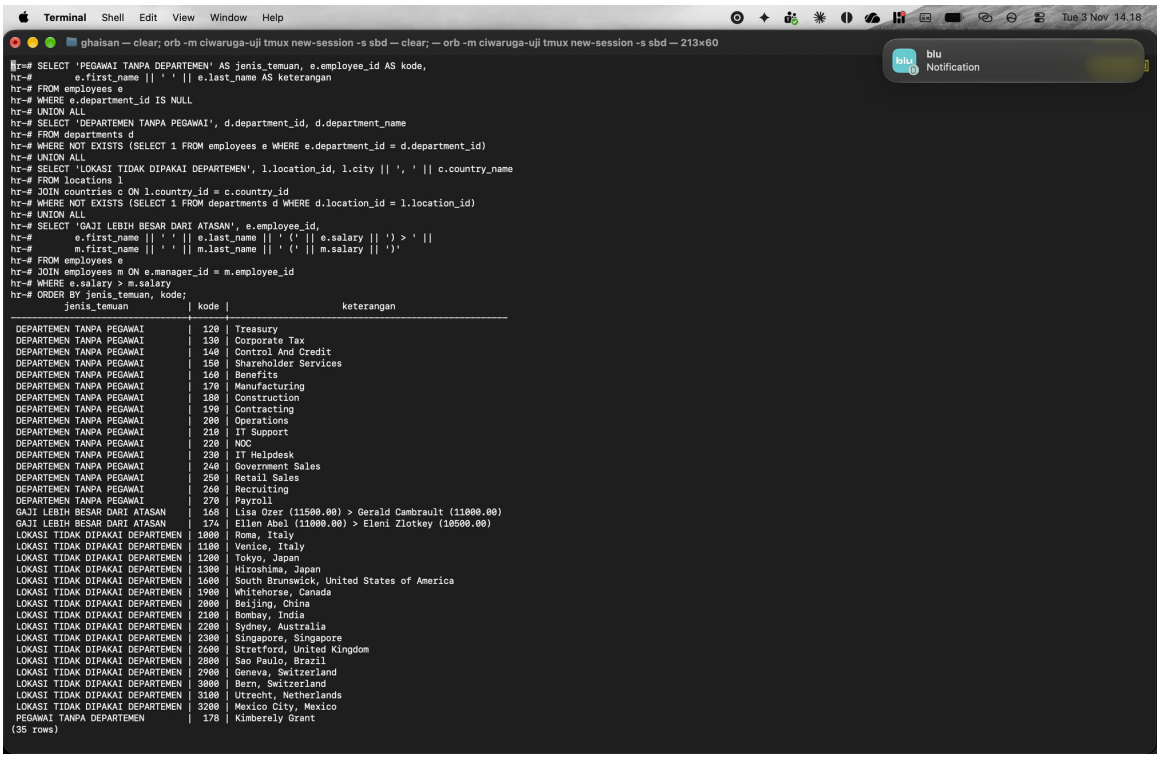
Syntax

```
SELECT 'PEGAWAI TANPA DEPARTEMEN' AS jenis_temuan, e.employee_id AS kode,
       e.first_name || ' ' || e.last_name AS keterangan
FROM employees e
WHERE e.department_id IS NULL
UNION ALL
SELECT 'DEPARTEMEN TANPA PEGAWAI', d.department_id, d.department_name
FROM departments d
WHERE NOT EXISTS (SELECT 1 FROM employees e WHERE e.department_id = d.department_id)
UNION ALL
SELECT 'LOKASI TIDAK DIPAKAI DEPARTEMEN', l.location_id, l.city || ', ' ||
    c.country_name
FROM locations l
JOIN countries c ON l.country_id = c.country_id
WHERE NOT EXISTS (SELECT 1 FROM departments d WHERE d.location_id = l.location_id)
UNION ALL
SELECT 'GAJI LEBIH BESAR DARI ATASAN', e.employee_id,
       e.first_name || ' ' || e.last_name || ' (' || e.salary || ') > ' ||
       m.first_name || ' ' || m.last_name || ' (' || m.salary || ')'
FROM employees e
JOIN employees m ON e.manager_id = m.employee_id
WHERE e.salary > m.salary
ORDER BY jenis_temuan, kode;
```

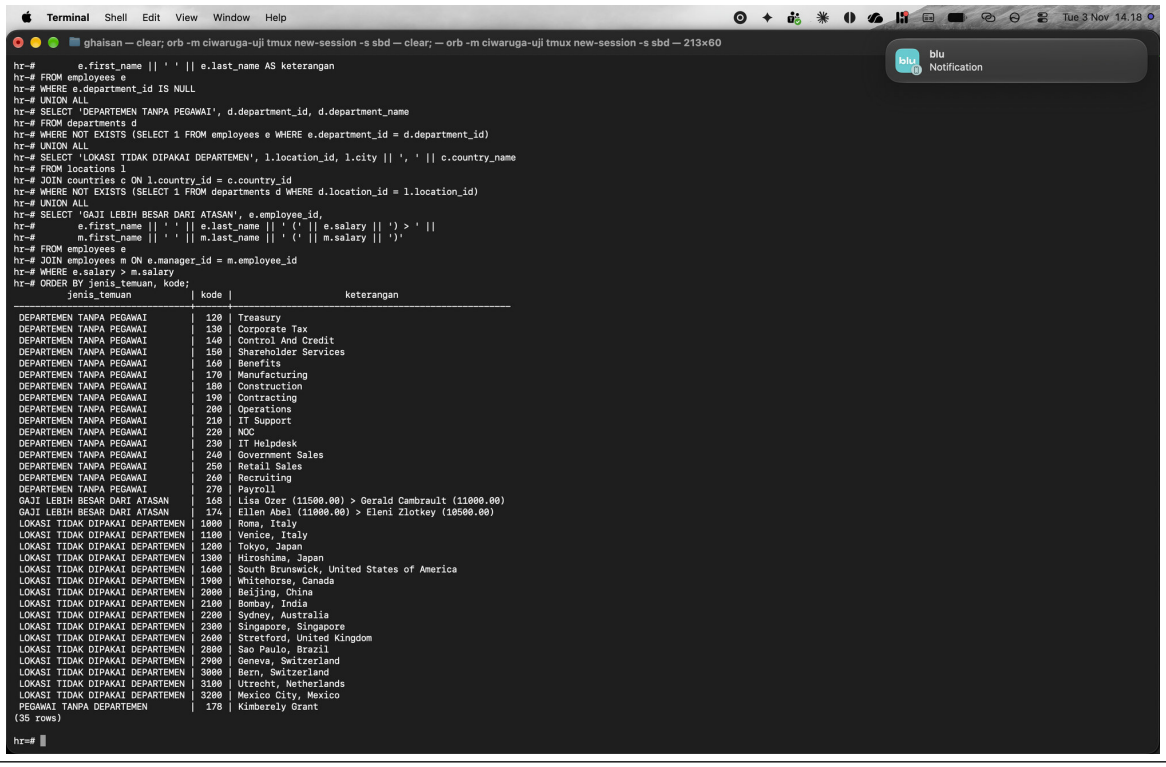
Keterangan

Empat query digabung dengan UNION ALL. Setiap query menghasilkan tiga kolom dengan tipe yang sama (teks, angka, teks), dan ORDER BY di akhir berlaku untuk seluruh hasil gabungan.
Pegawai tanpa departemen dicari dengan department_id IS NULL. Departemen tanpa pegawai dan lokasi yang tidak dipakai departemen dicari dengan NOT EXISTS. Gaji lebih besar dari atasan dicari dengan self join dan e.salary > m.salary, keterangannya disusun dengan operator ||.
Hasilnya 35 temuan: 16 departemen tanpa pegawai (ID 120 sampai 270), 2 pegawai bergaji lebih besar dari atasannya (Lisa Ozer 11500 > Gerald Cambrault 11000 dan Ellen Abel 11000 > Eleni Zlotkey 10500), 16 lokasi tidak dipakai departemen, dan 1 pegawai tanpa departemen (Kimberely Grant, ID 178).
Output lebih panjang dari satu layar, jadi ada dua screenshot: syntax dan awal output, lalu bagian akhir output.

Hasil Eksekusi (Screenshot)



```
hr=# SELECT 'PEGAWAI TANPA DEPARTEMEN' AS jenis_temuan, e.employee_id AS kode,
hr-#        e.first_name || ' ' || e.last_name AS keterangan
hr-# FROM employees e
hr-# WHERE e.department_id IS NULL
hr-# UNION ALL
hr-# SELECT 'DEPARTEMEN TANPA PEGAWAI', d.department_id, d.department_name
hr-# FROM departments d
hr-# WHERE NOT EXISTS (SELECT 1 FROM employees e WHERE e.department_id = d.department_id)
hr-# UNION ALL
hr-# SELECT 'LOKASI TIDAK DIPAKAI DEPARTEMEN', l.location_id, l.city || ', ' || c.country_name
hr-# FROM locations l
hr-# JOIN countries c ON l.country_id = c.country_id
hr-# WHERE NOT EXISTS (SELECT 1 FROM departments d WHERE d.location_id = l.location_id)
hr-# UNION ALL
hr-# SELECT 'GAJI LEBIH BESAR DARI ATASAN', e.employee_id,
hr-#        e.first_name || ' ' || e.last_name || ' (' || e.salary || ') > ' ||
hr-#        m.first_name || ' ' || m.last_name || ' (' || m.salary || ')'
hr-# FROM employees e
hr-# JOIN employees m ON e.manager_id = m.employee_id
hr-# WHERE e.salary > m.salary
hr-# ORDER BY jenis_temuan, kode;
         jenis_temuan           | kode |                      keterangan
--------------------------------+------+-------------------------------------------------------
 DEPARTEMEN TANPA PEGAWAI       |  120 | Treasury
 DEPARTEMEN TANPA PEGAWAI       |  130 | Corporate Tax
 DEPARTEMEN TANPA PEGAWAI       |  140 | Control And Credit
 DEPARTEMEN TANPA PEGAWAI       |  150 | Shareholder Services
 DEPARTEMEN TANPA PEGAWAI       |  160 | Benefits
 DEPARTEMEN TANPA PEGAWAI       |  170 | Manufacturing
 DEPARTEMEN TANPA PEGAWAI       |  180 | Construction
 DEPARTEMEN TANPA PEGAWAI       |  190 | Contracting
 DEPARTEMEN TANPA PEGAWAI       |  200 | Operations
 DEPARTEMEN TANPA PEGAWAI       |  210 | IT Support
 DEPARTEMEN TANPA PEGAWAI       |  220 | NOC
 DEPARTEMEN TANPA PEGAWAI       |  230 | IT Helpdesk
 DEPARTEMEN TANPA PEGAWAI       |  240 | Government Sales
 DEPARTEMEN TANPA PEGAWAI       |  250 | Retail Sales
 DEPARTEMEN TANPA PEGAWAI       |  260 | Recruiting
 DEPARTEMEN TANPA PEGAWAI       |  270 | Payroll
 GAJI LEBIH BESAR DARI ATASAN   |  168 | Lisa Ozer (11500.00) > Gerald Cambrault (11000.00)
 GAJI LEBIH BESAR DARI ATASAN   |  174 | Ellen Abel (11000.00) > Eleni Zlotkey (10500.00)
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 1000 | Roma, Italy
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 1100 | Venice, Italy
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 1200 | Tokyo, Japan
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 1300 | Hiroshima, Japan
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 1600 | South Brunswick, United States of America
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 1900 | Whitehorse, Canada
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 2000 | Beijing, China
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 2100 | Bombay, India
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 2200 | Sydney, Australia
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 2300 | Singapore, Singapore
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 2600 | Stretford, United Kingdom
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 2800 | Sao Paulo, Brazil
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 2900 | Geneva, Switzerland
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 3000 | Bern, Switzerland
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 3100 | Utrecht, Netherlands
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 3200 | Mexico City, Mexico
 PEGAWAI TANPA DEPARTEMEN       |  178 | Kimberely Grant
(35 rows)
```



```
hr-#        e.first_name || ' ' || e.last_name AS keterangan
hr-# FROM employees e
hr-# WHERE e.department_id IS NULL
hr-# UNION ALL
hr-# SELECT 'DEPARTEMEN TANPA PEGAWAI', d.department_id, d.department_name
hr-# FROM departments d
hr-# WHERE NOT EXISTS (SELECT 1 FROM employees e WHERE e.department_id = d.department_id)
hr-# UNION ALL
hr-# SELECT 'LOKASI TIDAK DIPAKAI DEPARTEMEN', l.location_id, l.city || ', ' || c.country_name
hr-# FROM locations l
hr-# JOIN countries c ON l.country_id = c.country_id
hr-# WHERE NOT EXISTS (SELECT 1 FROM departments d WHERE d.location_id = l.location_id)
hr-# UNION ALL
hr-# SELECT 'GAJI LEBIH BESAR DARI ATASAN', e.employee_id,
hr-#        e.first_name || ' ' || e.last_name || ' (' || e.salary || ') > ' ||
hr-#        m.first_name || ' ' || m.last_name || ' (' || m.salary || ')'
hr-# FROM employees e
hr-# JOIN employees m ON e.manager_id = m.employee_id
hr-# WHERE e.salary > m.salary
hr-# ORDER BY jenis_temuan, kode;
         jenis_temuan           | kode |                      keterangan
--------------------------------+------+-------------------------------------------------------
 DEPARTEMEN TANPA PEGAWAI       |  120 | Treasury
 DEPARTEMEN TANPA PEGAWAI       |  130 | Corporate Tax
 DEPARTEMEN TANPA PEGAWAI       |  140 | Control And Credit
 DEPARTEMEN TANPA PEGAWAI       |  150 | Shareholder Services
 DEPARTEMEN TANPA PEGAWAI       |  160 | Benefits
 DEPARTEMEN TANPA PEGAWAI       |  170 | Manufacturing
 DEPARTEMEN TANPA PEGAWAI       |  180 | Construction
 DEPARTEMEN TANPA PEGAWAI       |  190 | Contracting
 DEPARTEMEN TANPA PEGAWAI       |  200 | Operations
 DEPARTEMEN TANPA PEGAWAI       |  210 | IT Support
 DEPARTEMEN TANPA PEGAWAI       |  220 | NOC
 DEPARTEMEN TANPA PEGAWAI       |  230 | IT Helpdesk
 DEPARTEMEN TANPA PEGAWAI       |  240 | Government Sales
 DEPARTEMEN TANPA PEGAWAI       |  250 | Retail Sales
 DEPARTEMEN TANPA PEGAWAI       |  260 | Recruiting
 DEPARTEMEN TANPA PEGAWAI       |  270 | Payroll
 GAJI LEBIH BESAR DARI ATASAN   |  168 | Lisa Ozer (11500.00) > Gerald Cambrault (11000.00)
 GAJI LEBIH BESAR DARI ATASAN   |  174 | Ellen Abel (11000.00) > Eleni Zlotkey (10500.00)
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 1000 | Roma, Italy
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 1100 | Venice, Italy
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 1200 | Tokyo, Japan
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 1300 | Hiroshima, Japan
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 1600 | South Brunswick, United States of America
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 1900 | Whitehorse, Canada
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 2000 | Beijing, China
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 2100 | Bombay, India
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 2200 | Sydney, Australia
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 2300 | Singapore, Singapore
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 2600 | Stretford, United Kingdom
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 2800 | Sao Paulo, Brazil
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 2900 | Geneva, Switzerland
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 3000 | Bern, Switzerland
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 3100 | Utrecht, Netherlands
 LOKASI TIDAK DIPAKAI DEPARTEMEN | 3200 | Mexico City, Mexico
 PEGAWAI TANPA DEPARTEMEN       |  178 | Kimberely Grant
(35 rows)

hr=#
```

Latihan No. 10

Soal

Manajemen ingin melihat bagaimana pegawai berpindah posisi dibanding posisinya sekarang.
Untuk setiap baris job_history, tampilkan:

- pegawai (nama lengkap), jabatan_lama, departemen_lama
- start_date, end_date
- lama_tahun: lama menjabat = (end_date − start_date) / 365,25, dibulatkan 1 desimal
- jabatan_sekarang dan departemen_sekarang (isi - jika tidak ada)
- jenis_perpindahan:
  - jabatan dan departemen lama sama dengan yang sekarang: Kembali ke posisi semula
  - hanya departemennya sama: Ganti jabatan, departemen sama
  - selain itu: Pindah departemen

Urutkan menurut employee_id, lalu start_date.

Syntax

```
SELECT e.first_name || ' ' || e.last_name AS pegawai,
       jl.job_title AS jabatan_lama, dl.department_name AS departemen_lama,
       jh.start_date::date AS start_date, jh.end_date::date AS end_date,
       ROUND((jh.end_date::date - jh.start_date::date) / 365.25, 1) AS lama_tahun,
       COALESCE(js.job_title, '-') AS jabatan_sekarang,
       COALESCE(ds.department_name, '-') AS departemen_sekarang,
       CASE
         WHEN jh.job_id = e.job_id AND jh.department_id = e.department_id THEN
    'Kembali ke posisi semula'
         WHEN jh.department_id = e.department_id THEN 'Ganti jabatan, departemen sama'
         ELSE 'Pindah departemen'
       END AS jenis_perpindahan
FROM job_history jh
JOIN employees e ON jh.employee_id = e.employee_id
JOIN jobs jl ON jh.job_id = jl.job_id
LEFT JOIN departments dl ON jh.department_id = dl.department_id
LEFT JOIN jobs js ON e.job_id = js.job_id
LEFT JOIN departments ds ON e.department_id = ds.department_id
ORDER BY jh.employee_id, jh.start_date;
```

Keterangan

Setiap baris job_history digabung dengan employees untuk nama dan posisi sekarang. Tabel jobs dan departments dipakai dua kali: alias jl dan dl untuk jabatan dan departemen lama, js dan ds untuk jabatan dan departemen sekarang (dengan COALESCE ke - kalau tidak ada).
start_date dan end_date bertipe timestamp, jadi diubah ke date dulu supaya hasil pengurangannya berupa jumlah hari, lalu dibagi 365,25 dan dibulatkan 1 desimal.
jenis_perpindahan ditentukan CASE yang dicek berurutan: job_id dan department_id lama sama dengan sekarang, lalu hanya department_id yang sama, selain itu Pindah departemen.
Hasilnya 10 baris: 6 Pindah departemen, 3 Ganti jabatan dengan departemen sama, dan 1 Kembali ke posisi semula (Jonathon Taylor yang pernah menjadi Sales Representative di Sales dan sekarang kembali di posisi itu). Jennifer Whalen tercatat Pindah departemen walaupun jabatannya sama, karena dulu di Executive dan sekarang di Administration.

Hasil Eksekusi (Screenshot)

